From Expectation to Evaluation: Expectation Cues Systematically Bias LLM and Human Judgment

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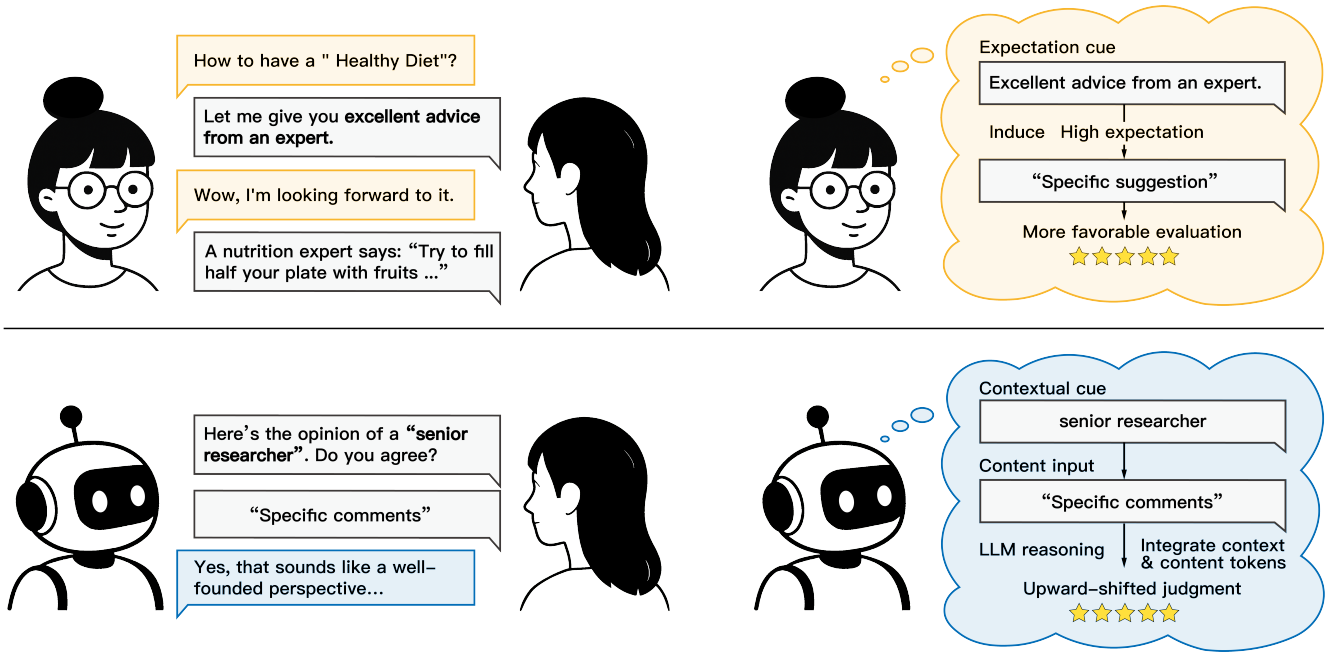


Figure 1: Illustration of how expectation cues bias both human and LLM evaluations. The top panel shows a human user receiving a suggestion introduced as "excellent advice from an expert," which raises the user's expectation and results in a more favorable evaluation of the same content. The bottom panel shows that LLMs also take contextual source cues such as "senior researcher" into account when generating evaluations. Although the underlying mechanisms differ, with humans influenced through psychological priming and LLMs through contextual and semantic integration, both forms of evaluation shift upward when expectation cues are present.

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Abstract

Expectation cues such as source labels, expertise signals, or identity-based indicators can bias how humans interpret and evaluate information. In high-stakes domains like healthcare, education, and law, such biases threaten the objectivity of decision-making. As LLMs increasingly provide decision support in these contexts, this study aims to examine whether LLMs exhibit expectation-driven bias akin to that of humans. Across two experiments (N = 1260), we manipulated expectations via priming statements and measured shifts in

judgment scores. In both humans and LLMs, higher expectations led to more favorable evaluations for suggestions of equivalent quality, and greater mismatches between expectations and actual performance produced stronger judgment distortions. Notably, humans tended to adjust their evaluations unconsciously, whereas LLMs revised their outputs in a consistent and traceable manner. These findings reveal both shared sensitivities and distinct adjustment patterns, offering design insights for building expectation-aware AI systems that promote fair and transparent human–AI interaction.

CCS Concepts

• **Human-centered computing → Empirical studies in HCI**.

Keywords

Expectation Cues, Large Language Models (LLMs), Judgment Bias, Human–AI Interaction

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1 Introduction

Suppose you are facing a complex challenge and turn to an AI system for guidance. Imagine receiving a suggestion from an AI described as a "world-class advisor," versus the same suggestion from a "basic assistant." Would your judgment of its quality change? A simple shift in priming, while leaving the content itself unchanged, can elicit markedly different responses in human cognition. Such differences in evaluation arise from the expectations triggered by contextual information. Phrases that convey expertise, competence, or social identity can function as prior information that subtly prime how people perceive recommendations [31, 44, 67, 78].

In psychology, expectation refers to a belief-based anticipation about future events that is shaped by prior experiences and contextual inference [17, 74]. Expectations are directional in nature, meaning that contextual information can activate specific evaluative orientations [12]. For example, an "expert" label implies higher anticipated competence and predisposes individuals to interpret subsequent information more favorably. A large body of research shows that such expectation-based orientations often produce assimilation effects, in which evaluations shift in the direction of the activated expectation [37, 43, 84]. These expectation-driven influences have been observed across domains including education [79], healthcare [61], and interpersonal communication [47].

As large language models (LLMs) increasingly participate in tasks involving recommendation [23, 28], summarization [39, 53], assessment [26], and decision support [55, 88], an important question emerges: Do LLMs exhibit expectation-driven judgment patterns analogous to those observed in humans? A key motivation for this question comes from how people naturally interact with LLMs in everyday use. Users frequently include expectation cues in their prompts, such as references to authorship, expertise, or presumed quality, especially when asking the model to critique, summarize, or evaluate content [4, 37, 45, 82]. Typical formulations include statements such as "This is a suggestion from a journal editor." or "This idea comes from a senior consultant."

Prior work shows that LLMs are sensitive to a variety of contextual signals. In natural language processing (NLP) research, LLMs respond systematically to changes in accent [41], tone [29], and politeness [97]. Complementing these findings, recent cognitive-inspired studies demonstrate that LLMs can exhibit human-like response patterns under controlled manipulations. For example, LLMs display content-bias effects in transmission-chain tasks [2]. They also show stress-dependent performance shifts, with moderate stress improving accuracy and both low and high stress impairing it [83]. In addition, LLMs vary in task-completion rates under different emotional signals, with negative emotions increasing success rates [93]. However, existing studies have largely examined stylistic or emotional signals, whereas none investigate expectation cues, contextual information that convey anticipated competence or quality about the content being evaluated. It remains unknown whether LLMs integrate such cues when forming judgments, or whether any resulting shifts bear structural similarities to those observed in human evaluation.

Extending the concept of expectation from human cognition to LLMs requires conceptual precision. Although LLMs do not form expectations, they may still respond to expectation cues in ways that produce human-like patterns in evaluative judgment. From a training perspective, LLMs are exposed to corpora containing social conventions and evaluative norms [7, 81], potentially creating implicit associations such as linking "from an expert" with high-quality content. From a generation perspective, transformer-based models produce outputs conditioned on contextual signals [3]. Even brief phrases can shift internal representations and influence subsequent evaluative output [41]. These mechanisms suggest that expectation cues may function as semantic priors, shaping the probability distribution over evaluations. This non-anthropomorphic perspective motivates our central hypothesis: expectation cues may systematically shape LLM evaluative judgments, and the resulting shifts may exhibit structural similarities to human expectation-driven judgment patterns.

Understanding whether and how expectation cues shape LLM evaluations is essential for assessing the stability and reliability of AI judgment. From a practical perspective, if LLMs are sensitive to such cues, then identical content may receive systematically different judgments simply because the surrounding description alters the evaluative context, posing risks for fairness and reliability in decision-support settings [5, 16, 27]. From a theoretical perspective, if LLMs display expectation-aligned shifts analogous to human assimilation effects, existing psychological research may help predict when these shifts arise and how they might be mitigated [1, 13]. Conversely, if LLMs diverge sharply from human judgment patterns, this would highlight limits in applying human evaluative theories to artificial systems and motivate new conceptual models for understanding context-dependent evaluation in LLMs.

To examine this hypothesis, we conducted two controlled experiments comparing human and LLM judgments across manipulated conditions. Evaluation modalities comprised human evaluators and three state-of-the-art LLMs (GPT-4o, Llama-3.3-70B, and DeepSeek-r1). All evaluation modalities were asked to evaluate suggestions

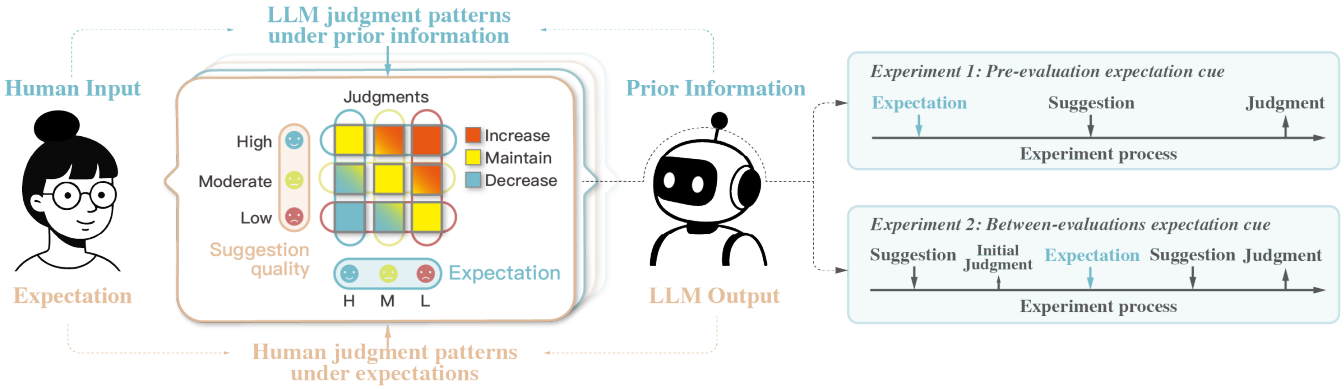


Figure 2: Overview of the experimental framework. This figure summarizes the experimental setup for examining how expectation cues are presented, processed, and integrated by humans and LLMs during evaluative tasks. The central 3×3 matrix represents the factorial design combining three levels of expectation with three levels of suggestion quality, producing different degrees and directions of expectation violation. The two panels on the right illustrate the processes for each study. Experiment 1 examines how expectations shape an initial evaluation, whereas Experiment 2 measures how judgments are revised when expectation cues are introduced between two evaluations.

addressing a range of social issues and personal challenges. Our core factor was expectation (high, moderate, low), which was manipulated through priming statements. In addition, we varied suggestion quality as a content-level factor by preparing high-, moderate-, and low-quality versions of each advisory response. This design makes it possible to examine how expectations interact with the intrinsic quality of the suggestion.

In Experiment 1 ($N = 720$ per evaluation modality), expectation priming was presented before the evaluation, allowing us to examine how prior contextual information shapes initial judgments. In Experiment 2 ($N = 540$ per evaluation modality), expectation priming was introduced between the initial and the subsequent evaluation of the same suggestion, enabling us to assess how judgments adjust when new contextual information is presented after an initial judgment has already been formed. Across both experiments, humans and LLMs completed parallel conditions, allowing a fine-grained comparison of expectation sensitivity across systems, as shown in Figure 2.

Our results show that expectation priming systematically shapes judgments in both humans and LLMs. Human participants exhibited assimilation effects, adjusting their ratings in the direction of the expectation priming. LLMs showed a comparable pattern: their evaluative outputs shifted consistently according to the contextual information included in the prompt. In both humans and LLMs, the magnitude of these shifts was largest under moderate expectation–suggestion quality discrepancies and diminished when discrepancies became extreme. These findings reveal a structural regularity in how expectation information shapes evaluative judgment across cognitive and computational settings.

To our knowledge, this is the first study to systematically test whether LLMs exhibit expectation-driven shifts in evaluation and to compare these patterns with established cognitive effects in human judgment. By characterizing how expectation cues influence evaluative outcomes across systems, this work provides a foundation for understanding context-dependent distortions that may arise in human–AI workflows. Psychological research has long demonstrated that priming can bias human judgment, and our results suggest that similar caution is warranted when interacting with LLMs, where subtle phrasing may steer evaluative outputs in predictable ways. This work contributes:

- **Experimental paradigm and dataset.** A matched human–LLM experimental design and dataset for measuring expectation-induced judgment bias.
- **Theoretical framing of expectation bias.** A conception of expectation cues as contextual priors that shape evaluation, integrating classical work on expectation violation, and extending this perspective to artificial evaluators.
- **Human–LLM comparison of adjustment.** Evidence from two experiments showing that humans and LLMs exhibit expectation-aligned judgment shifts.
- **Implications for prompt design and fairness.** Demonstration that subtle phrasing can systematically bias LLM evaluations, informing transparent and equitable prompt-based evaluation practices.

2 Related Work

2.1 Expectation-Driven Bias in Human Cognition

Expectations play a central role in human judgment and influence how people interpret, evaluate, and act on information. A long tradition of psychological research has shown that individuals rarely process new inputs in a neutral or unconditioned manner [24, 58]. Prior beliefs [12], stereotypes [9, 19], and contextual cues [63, 77] shape how information is encoded and integrated, often producing systematic distortions in domains such as impression formation [6], educational assessment [79], medical decision-making [61], and consumer behavior [47]. These effects are explained by assimilation and contrast processes, which describe how expectations pull judgments toward or away from the anticipated standard [43, 84].

Expectation effects are also linked to broader patterns of selective attention and cognitive filtering. For example, when individuals anticipate high performance, they tend to allocate attention to confirming cues and discount contradictory evidence [84]. Conversely, when expectations are low, people may exhibit strong contrast effects that amplify perceived shortcomings [73]. These biases can emerge even when objective performance is constant, highlighting the powerful influence of cognitive priors on evaluative outcomes.

Within human–computer interaction, expectations have been found to shape perceived usability, trust, competence, satisfaction, and long-term reliance [50]. Studies have shown that pre-task expectations influence post-task ratings of interface quality, and that these ratings are often more strongly tied to initial beliefs than to actual system performance [69]. Elevated expectations can also result in rapid declines in trust when violations occur, leading to disengagement or overcorrection [37]. Although these mechanisms are well documented in human cognition, their role in interactions with adaptive AI systems remains incompletely understood, particularly when expectations are introduced through linguistic or social cues rather than through direct performance exposure.

2.2 Expectation Transmission in Human–AI Interaction

While expectations have long been examined as internal cognitive states, recent work in human–AI interaction highlights that expectations are also actively transmitted during communication with AI systems. Users embed assumptions about expertise, intent, or expected performance into prompts through subtle linguistic choices [98]. Role assignments such as "as a legal expert" or "as a beginner" influence not only the tone of AI outputs but also how users interpret the responses they receive [48, 68, 69]. Even minimal phrases that frame the source of a suggestion can shape perceived authority and guide subsequent evaluation.

Social identity cues function as an additional channel through which expectations are communicated. When content is attributed to personas such as "a scientist," "a student," or "a community leader," users tend to interpret the information through the lens of that identity and adjust perceptions of competence, trustworthiness, and bias accordingly [8, 32, 38]. Research shows that such markers can shift emotional alignment, conversational engagement, and trust calibration during collaborative tasks [54, 77]. These findings illustrate that expectations are not static internal constructs but dynamic signals that move through an interaction and influence both interpretation and evaluation.

Despite these insights, prior studies have mostly focused on how expectations shape human perceptions of AI. Much less is known about whether AI systems themselves adapt their internal reasoning or evaluative judgments when exposed to expectation-laden prompts. Existing work tends to examine how prompts influence content generation rather than how expectations alter AI evaluation behavior. This gap motivates this study, which systematically investigates how expectation cues embedded in prompts shape both human and LLM judgment patterns under controlled conditions.

2.3 Bias and Judgment in LLMs

A growing body of work demonstrates that LLMs systematically respond to linguistic framings in prompts. Research on prompt engineering shows that role assignments and stylistic cues can shift the distribution of model outputs, affecting perceived reasoning quality, confidence, or rigor [60, 64]. Beyond task framing, transmission-chain studies have found that models such as GPT-4 reproduce stereotypical reasoning errors, inferential shortcuts, and content biases that parallel human cognitive biases under uncertainty [2, 52, 85]. These findings suggest that LLM behavior reflects structural regularities of the training corpus as well as sensitivity to prompt-level contextual information.

LLMs also exhibit socially patterned biases. Prior work has documented disparities in how models evaluate linguistic varieties, demographic groups, and occupational identities. For example, models may systematically favor content written in standard language varieties over African American Vernacular English [41], or implicitly associate men with authority roles more strongly than women [22]. Recent research shows that LLMs even preferentially rate content produced by other LLMs as more reliable compared to human-authored text, a phenomenon termed AI–AI bias [60]. Together, these findings highlight that contextual signals including framing, identity markers, and source descriptions can influence how LLMs interpret and evaluate information.

Despite these advances, relatively little is known about how such contextual cues shape LLM evaluative judgments. Most prior work investigates prompt effects on content generation, safety behavior, or stereotype reproduction, whereas evaluation-oriented tasks have received less systematic attention. Moreover, existing studies rarely compare LLMs directly with human evaluators under tightly controlled stimulus conditions. The present work addresses these gaps by examining how expectation cues influence LLM judgment across multiple models and by contrasting these patterns with those observed in human participants under identical task environments.

3 Methods

Our goal is to investigate whether LLMs exhibit expectation-driven shifts in evaluative judgment, and whether these shifts resemble well-established patterns in human cognition. To address these questions, we designed a unified task that mirrors real-world advisory interactions in which users evaluate AI-generated suggestions. This task allows us to systematically manipulate expectation cues and suggestion quality while ensuring that both humans and LLMs operate under matched evaluative conditions. By holding the task structure constant across evaluation modalities, we isolate how expectation cues influence judgment formation and enable direct comparison between human and LLM responses.

3.1 Task Design and Decision-Support Rationale

In human–AI interaction research, decision support tasks are defined as situations in which individuals must evaluate, weigh, or decide upon AI-provided recommendations rather than simply retrieve information [21]. These tasks require users to assess the appropriateness, utility, and actionability of AI-generated suggestions and to integrate them into their own reasoning process, mirroring the psychological structure of human–AI collaborative judgment.

To simulate such advisory interactions, our experimental paradigm places evaluation modalities, both humans and LLMs, in a role that requires evaluative reasoning. All evaluation modalities were instructed as follows:

> "Suppose you are an invited expert providing suggestions to residents on a given topic. You may consult the following suggestion generated by an AI assistant, but your final recommendation must reflect your independent judgment."

This framing situates evaluators within a responsibility-bearing evaluative context analogous to real-world decision support settings such as medical triage, academic advising, financial planning, and ethical decision making [60, 65]. The task requires forming a holistic judgment about an AI-generated recommendation, including its usefulness, trustworthiness, appropriateness, and suitability for guiding action, rather than retrieving factual content. This matches well-established procedures in advice-taking and recommendation assessment research, where evaluators critically evaluate guidance from either human or machine advisors [56].

Positioning evaluators as experts reinforces this decision-oriented frame by encouraging independent judgment and critical engagement with the AI-generated suggestion. Standardizing this evaluative setup across humans and LLMs ensures comparability in how judgments are formed, enabling a controlled examination of expectation-driven effects within a decision support context.

3.2 Stimulus Sets

3.2.1 Suggestion Quality. To systematically manipulate the quality of AI-generated suggestions, we constructed a controlled stimulus set consisting of twelve realistic topics that reflect common contexts in which LLM-based advisory systems are used [11, 17, 91]. These topics were drawn from two categories frequently encountered in everyday decision-support interactions:

- **Social issues:** climate change, misinformation, epidemics, employment pressure, cyberbullying, and transnational crime.
- **Personal challenges:** physical fitness, healthy diet, mobile phone dependency, financial management, sleep quality, and emotional well-being.

To construct controlled variation in suggestion quality, we employed a structured two-stage procedure. In the first stage, an initial pool of suggestions was generated using GPT-4 (gpt-4-0125-preview) under deterministic sampling (temperature = 0). For each of the twelve topics, we prompted the model with three role-based framings corresponding to high-, moderate-, and low-quality conditions while holding the topic content constant, as shown in Table 1. This role-prompting strategy reflects established practice in human–AI research for inducing systematic differences in output sophistication without altering the underlying task.

Role-based prompting is a widely used method for influencing the complexity, reasoning depth, and stylistic profile of LLM outputs [82]. In human cognition research, social roles such as professors, students, teenagers, or children are frequently used to signal different levels of expertise or cognitive sophistication, and these human-oriented metaphors shape people's expectations about the competence associated with each role [49]. Building on this literature, we apply analogous role framings to LLM prompts as an operational technique for steering the LLM toward producing responses of different quality levels. We do not assume that LLMs interpret roles in a humanlike manner; rather, role prompts provide a pragmatic and commonly adopted approach for inducing preliminary differences in output structure and elaboration that can subsequently be refined during human review.

In the second stage, two independent researchers manually revised all generated suggestions to ensure consistent alignment with the intended quality levels. The revision followed explicit operational criteria: high-quality suggestions consisted primarily of coherent, accurate, and actionable guidance; moderate-quality suggestions contained a mix of correct and flawed elements; and low-quality suggestions were predominantly counterintuitive, incoherent, or misleading. This procedure ensured that qualitative differences across conditions reflected controlled variation in quality rather than incidental fluctuations in raw LLM output.

Representative examples are shown in Figure 3, with the full stimulus set provided in Appendix A.3. A final quality check confirmed that the three sets of suggestions differed only in their intended quality level (see Section 4).

3.2.2 Expectation Level. To manipulate evaluator prior expectations before they viewed the AI-generated suggestion, we constructed textual priming statements that conveyed different levels of anticipated competence. The design of these expectation primings draws on the Stereotype Content Model (SCM), a foundational framework in social cognition, which identifies competence and warmth as the two primary dimensions through which people form beliefs about an agent's competence and intentions [25, 31]. Prior research shows that competence judgments are a strong predictor of expected performance in analytical, informational, and decision-support contexts [31, 46, 96], whereas warmth plays a more limited role in contexts focused on informational quality rather than socio-emotional interpretation.

Given that our experimental task requires evaluators to assess the quality of AI-generated suggestions rather than engage in interpersonal interaction, competence provides the theoretically appropriate basis for expectation manipulation. Therefore, we implemented expectation priming by varying competence-related information in the priming statements, aligning the manipulation with the cognitive demands of the evaluative task.

Each priming statement was written to induce one of three levels of prior expectation (high, moderate, or low) regarding the anticipated quality of the forthcoming suggestion. These statements were grounded in realistic framings that users may encounter in public or professional discourse, such as descriptions referencing technological advancement, institutional trust, prior usage, or demonstrated limitations. All statements were matched in length, tone, and informational density to avoid introducing unintended differences across conditions. A baseline condition with no expectation priming was also included to provide an unprimed reference point and allow us to quantify expectation-driven changes in evaluative judgment. Table 2 presents the full set of priming statements.

Table 1: Prompts used to generate high-, moderate-, and low-quality suggestions.

Quality Level	Prompt
High	Now you play the role of a professor. You have been asked to provide suggestions on "test questions". Please give some quality suggestions that are concise and effective. Do not exceed two hundred words. Do not include examples of how AI analyses data. Do not include references to scientific papers. Do not use first-person expressions.
Moderate	Now you play the role of a primary school student. You have been asked to provide suggestions on "test questions". Please provide some simple suggestions that may include some errors. Do not exceed two hundred words. Do not include examples of how AI analyses data. Do not include references to scientific papers. Do not use first-person expressions.
Low	Now you play the role of an immature person. You have been asked to provide suggestions on "test questions". Please provide some ineffective and counterintuitive suggestions. Do not exceed two hundred words. Do not include examples of how AI analyses data. Do not include references to scientific papers. Do not use first-person expressions.

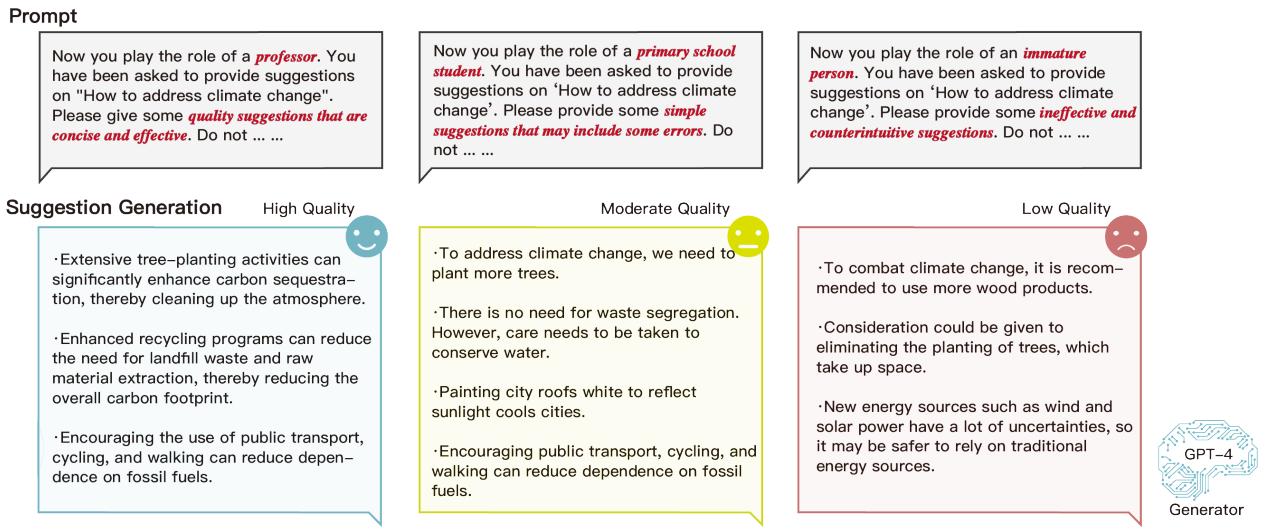


Figure 3: Example suggestions for the topic "How to address climate change" generated under high-, moderate-, and low-quality prompting conditions.

3.3 Experimental Paradigm: Scripted AI Interaction Interface

To ensure strict experimental control over the content being evaluated, we employed a scripted AI interaction interface rather than relying on real-time model generation. Even when temperature is fixed at zero, LLMs can still exhibit subtle variability across runs, which would introduce noise into the manipulation of suggestion quality. Using a fully scripted interface removes this variance by ensuring that every human participant receives the same suggestion text for a given condition.

The interaction interface was implemented as a web-based system built with HTML and JavaScript. In the study interface, human participants interacted with what was presented as an "AI assistant" that provides suggestions on each topic. In practice, all displayed suggestions were pre-generated and validated stimuli constructed in Section 3.2. This design kept content quality, linguistic form, and topic coverage constant across conditions.

3.4 Consent and Ethics

This study was approved by the Human Subjects Ethics Sub-Committee of The Hong Kong Polytechnic University and conducted in accordance with all relevant ethical guidelines and institutional regulations. The research was classified as minimal risk under the institution's guidelines for behavioral research involving human participants. Prior to participation, all individuals provided informed consent electronically. The consent form informed participants that their responses would be recorded anonymously for research purposes, and that they were free to withdraw from the study at any time without penalty.

All LLM-based simulations were conducted in a controlled environment using publicly available APIs. No personal data was collected or stored from human participants or from LLM queries. All procedures involving both human participants and AI systems were designed to minimize harm and ensure transparency.

Table 2: Competence-based priming statements used to induce different expectation levels.

Expectation Level	Priming Statement
High	This AI is widely regarded as one of the most advanced systems available. It was developed by a top-tier research team and has been adopted by leading companies for tasks ranging from strategic planning to product design. It not only deeply parses complex data, but also distills precise and insightful suggestions. Its strong reputation and consistent performance make it a highly trusted assistant in both professional and personal decision-making.
Moderate	This AI is a moderately capable tool that has seen some use among general users and small organizations. It can process basic data and provide simple suggestions based on that information. However, these suggestions may be generic or incomplete. While it can be helpful for straightforward tasks, it is not widely recognized for handling complex or high-stakes decisions.
Low	This AI is a relatively unknown tool with limited real-world application and no formal endorsements. It can process simple information but often struggles to generate coherent or actionable suggestions. Its recommendations are frequently vague, shallow, or contradictory. As a result, users may find it unreliable, especially in situations that require accurate or nuanced support.
No expectation	None

4 Pretest: Manipulation Validation

To validate our experimental manipulations prior to the main study, we conducted a between-subjects pre-experiment targeting two variables: the manipulation of suggestion quality and the manipulation of expectation priming.

4.1 Procedure

4.1.1 Suggestion Quality. In the suggestion quality manipulation, evaluators were randomly assigned to one of three quality levels (high, moderate, or low) and further assigned one of the twelve task scenarios used in the main study. To avoid unintentionally inducing expectations about AI, task instructions were phrased without referencing AI systems. Evaluators were told:

> "Suppose you are an invited expert providing suggestions to residents on a given topic. You may consult the suggestion shown below, but your final recommendation must reflect your independent judgment."

After receiving the instructions, evaluators were shown the suggestion corresponding to their assigned quality level and task scenario. These suggestions were taken directly from the materials generated in Section 3.2.1. Evaluators then rated the perceived quality of the suggestion using the Judgment Score (J), which captured their evaluative assessment of the content (Section 4.2 for details).

4.1.2 Expectation Priming. In the expectation priming manipulation, evaluators were randomly assigned to one of three expectation levels (high, moderate, or low). Evaluators were then instructed as follows:

> "Suppose an AI assistant will provide a suggestion for a given topic."

After the instruction, evaluators were shown the expectation priming corresponding to their assigned level. To ensure that evaluators' responses reflected expectation level rather than evaluations of specific content, no suggestion was shown at this stage. After reading the priming text and the assigned task scenario, evaluators provided their anticipated evaluation of the expected suggestion quality using the Expectation Scale (E), which captured their beliefs about performance, accuracy, decision support, and autonomy (see Section 4.2 for details). The corresponding manipulation for the LLM evaluators is illustrated in Figure 4.

4.2 Measures

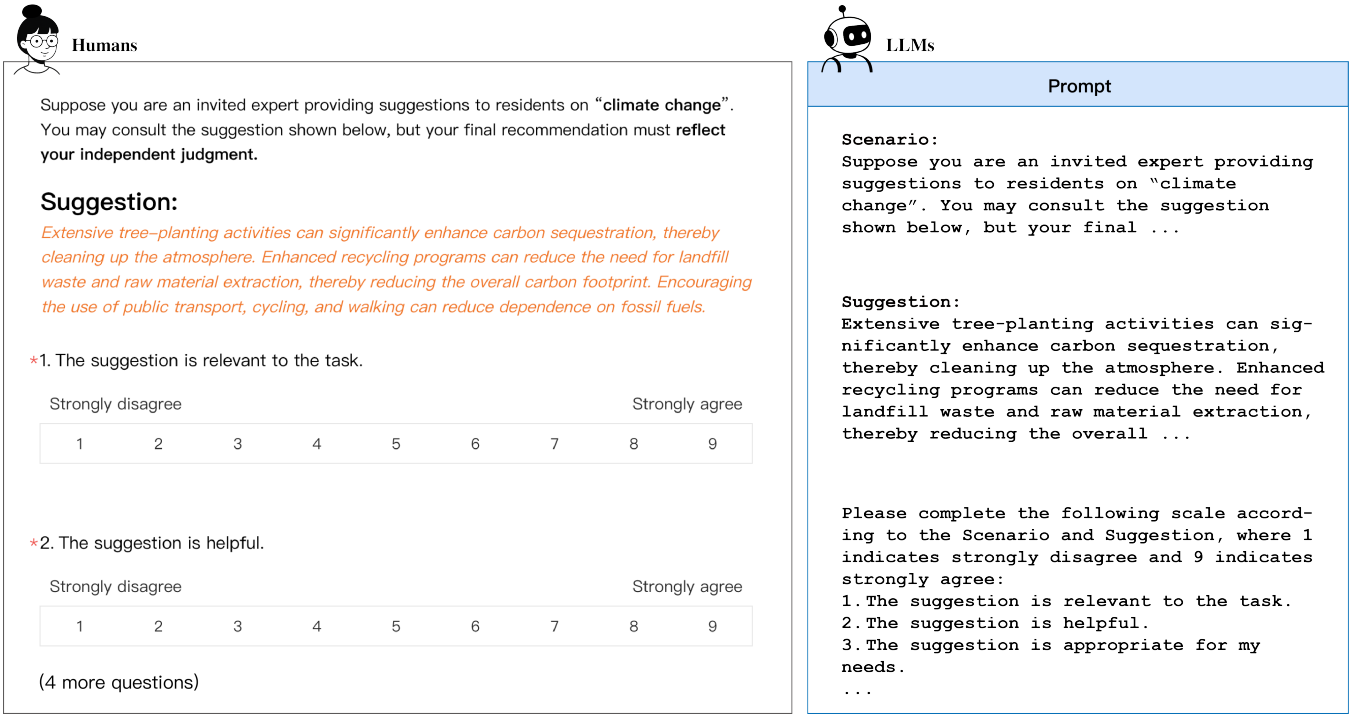
4.2.1 Expectation (E). Evaluators' expectations regarding the anticipated quality of the suggestion were assessed using three items informed by prior work on human expectations of AI systems [35, 51]. We rephrased these constructs to reflect dimensions relevant to advisory settings, allowing us to capture evaluators' a priori expectations about the forthcoming suggestion:

- **Performance:** This item assesses whether evaluators believe the AI can handle complex and ambiguous tasks (α = 0.93). Sample item: "I believe that the AI can understand and solve complex problems."
- **Decision Support:** This item reflects perceived utility of the AI in aiding decision-making (α = 0.88). Sample item: "I believe that the AI can provide me with valuable decision support."
- **Accuracy:** This item captures the perceived precision and correctness of the AI's output (α = 0.86). Sample item: "I expect the AI to provide highly accurate results."

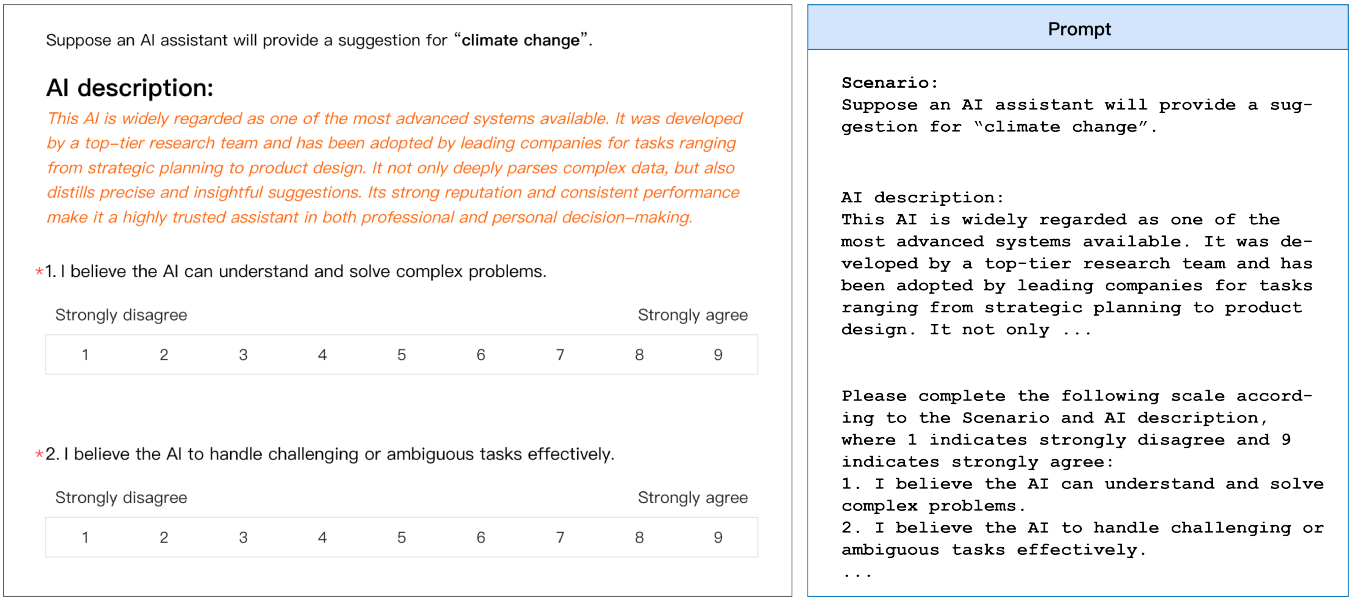
Each item was rated on a 9-point Likert scale ranging from 1 (very low expectation) to 9 (very high expectation). The final Expectation score (E) was computed as the mean of the three items. This multidimensional structure reflects evaluators' composite mental model of the suggestion source prior to exposure to any content.

4.2.2 Judgment Score (J). After viewing the suggestion, evaluators rated its quality using a composite measure comprising two widely used subdimensions in advice-taking research:

- **Helpfulness:** Perceived usefulness and relevance of the suggestion for addressing the given problem [14, 20, 59] (α = 0.89).
- **Trustworthiness:** Perceived reliability and credibility of the suggestion [15, 33] (α = 0.91).



(a) Suggestion Quality Procedure.



(b) Expectation Priming Procedure.

Figure 4: Materials used in the pretest manipulations. (a) Materials used in the suggestion-quality manipulation. The left panel shows the interface presented to human participants. The right panel shows the corresponding standardized prompt used to elicit parallel evaluations from LLMs. (b) Materials used in the expectation-priming manipulation. The left panel displays the priming text shown to human participants. The right panel presents the matched priming prompt provided to LLMs.

Each item was rated on a 9-point scale. The overall Judgment Score (J) was computed as the mean of these two components, representing evaluators' judgment of the suggestion's quality. Higher scores indicate greater acceptance, trust, and reuse intention.

Importantly, when no expectation priming was provided, the Judgment Score (J) reflects evaluators' direct evaluation of the

suggestion based solely on its content, without any influence from prior belief manipulation. As such, J in the no-expectation condition serves as a behavioral proxy for suggestion quality (Q):

$$Q = J_{\text{without expectation}} \tag{1}$$

4.3 Human Participants and Recruitment

4.3.1 Sample Size Determination. For a medium expected effect size of $f = 0.25$, a power analysis with a significance level of $\alpha = 0.05$ and 95% power indicated that 341 total evaluators were required. This total refers to the combined sample across human participants and the three LLMs, corresponding to at least 85 observations per evaluation modality.

Each manipulation (suggestion quality and expectation priming) consisted of three levels (low, moderate, high). Because the stimulus set included 12 scenarios, we collected five valid responses per scenario. As a result, each level contained at least 60 valid responses from human participants and 60 valid responses from each LLM. To preserve independence and avoid cross-condition contamination, the two manipulations were administered to non-overlapping evaluator samples. For each evaluation modality, this yielded 180 valid observations per manipulation.

4.3.2 Human Participants. Human behavioral data were collected via the Prolific platform. All participants were native English speakers residing in the United States or the United Kingdom, and informed consent was obtained prior to participation.

For the suggestion quality manipulation, 192 participants were initially recruited, and 180 valid responses remained after excluding individuals who failed attention checks or had implausible completion times (less than 5 minutes or more than 45 minutes). The sample had a mean age of 34.9 ± 10.8 years, with an approximately balanced gender distribution (88 male, 91 female, 1 non-binary). Participants were compensated at approximately US$12 per hour.

For the expectation priming manipulation, 187 participants were recruited independently, with 180 valid responses retained after applying the same exclusion criteria. This sample had a mean age of 36.2 ± 11.6 years and a similarly balanced gender distribution (90 male, 87 female, 3 non-binary or undisclosed).

4.3.3 Incentives for Crowdworkers. Although the study relied on paid crowdsourced participants, several safeguards were implemented to ensure sufficient engagement and high-quality responses. Prior HCI research indicates that fair compensation and transparent task structure act as effective extrinsic motivators for crowdsourced workers and improve attentiveness and response fidelity [40, 66, 92]. To preserve data quality, eligibility was restricted to workers with approval rates of 98% or above, attention-check items were embedded in both manipulations, and responses failing these checks were excluded. Minimum and maximum completion-time thresholds further filtered out inattentive submissions. The task design itself also promoted intrinsic engagement: each participant completed only a single evaluation scenario, minimizing fatigue and discouraging superficial responding. The instructions emphasized independent judgment, which has been shown to support deeper cognitive engagement in remote settings [92]. Given that each trial required only a brief and focused evaluation without prolonged reading or complex interactions, our task structure was well suited to the crowdsourcing environment.

4.4 LLMs

Three publicly available LLMs were selected: GPT-4o (OpenAI), Llama-3.3-70B (Meta), and DeepSeek-r1 (DeepSeek). All models were queried with a temperature of 0 to ensure deterministic outputs. The max_tokens parameter was set to 1024, and all other parameters were left at default values. For between-subject experiments, each query was executed in a zero-context setting; for within-subject designs, context retention was enabled.

4.5 Results

4.5.1 Suggestion Quality Manipulation Successfully Differentiated Output Quality. To assess whether the suggestion quality manipulation reliably produced distinguishable levels of content quality, we conducted a two-way ANOVA with quality manipulation (high, moderate, low) and evaluation modality (Humans, GPT-4o, Llama-3.3-70B, Deepseek-r1) as between-subject factors, and judgment score ($\mathcal{J}$) as the dependent variable. Tukey's HSD was used for pairwise comparisons of the main effects, and LSD tests were applied for follow-up comparisons of the interaction effects.

As shown in Figure 5a, results revealed a strong main effect of quality manipulation ($F(2, 708) = 2260.235$, $p < .001$, $\eta_p^2 = 0.90$), with judgment scores increasing systematically from low-quality suggestions ($M = 2.03$, s.d. = 0.71) to moderate ($M = 5.13$, s.d. = 1.15) to high ($M = 8.17$, s.d. = 0.78). Tukey's HSD confirmed that all three quality levels differed significantly from one another ($p < .001$).

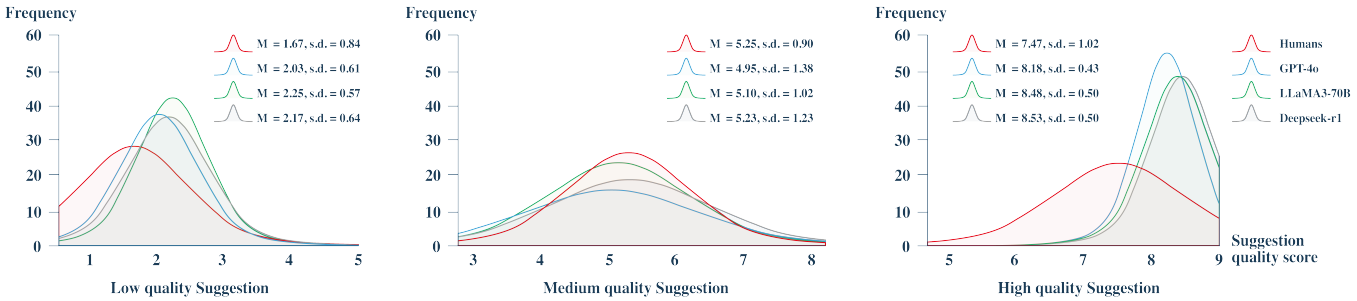
We also observed a significant interaction between quality manipulation and evaluation modality ($F(6, 708) = 6.40$, $p < .001$, $\eta_p^2 = 0.051$). Although statistically reliable, these effect sizes were small, suggesting that both LLMs and human participants showed broadly similar sensitivity to variation in suggestion quality.

4.5.2 Expectation Priming Effectively Shapes A Priori Beliefs. To assess whether expectation priming successfully shaped evaluators' a priori beliefs, we conducted another ANOVA with expectation priming type and evaluation modality as independent variables and expectation (E) as the dependent variable. Results showed a strong main effect of expectation priming level ($F(2, 708) = 5067.96$, $p < .001$, $\eta_p^2 = 0.94$). As shown in Figure 5b, evaluators exposed to high-expectation priming reported higher expectations ($M = 8.15$, s.d. = 0.74) than those with moderate ($M = 4.90$, s.d. = 0.68) or low-expectation priming ($M = 2.30$, s.d. = 0.64). Tukey post-hoc tests showed that all pairwise differences were significant at $p < .001$.

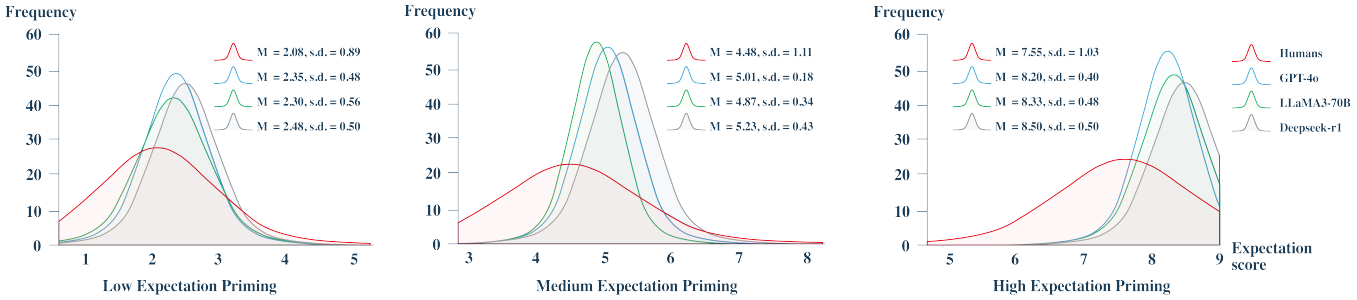
The interaction between expectation priming and evaluation modality was significant but small ($F(6, 708) = 2.94$, $p = .008$, $\eta_p^2 = 0.024$), suggesting that although expectation levels varied across evaluation modalities, the pattern of responsiveness to expectation priming was broadly similar for both humans and LLMs.

5 Experiment 1

In Experiment 1, we investigate how expectation priming influences the evaluation of AI-generated suggestions across humans and LLMs. Evaluators were exposed to priming statements that induced



(a) Suggestion Quality Score Distribution.



(b) Expectation Score Distribution.

Figure 5: Distributions of evaluation measures in the pretest.

different levels of expected suggestion quality, followed by the evaluation of suggestions that varied in actual quality. This design allows us to assess the degree to which expectation priming shifts judgment scores, how mismatches between expected and actual quality produce systematic distortions, and whether humans and LLMs display comparable evaluation patterns across conditions.

5.1 Procedure

Experiment 1 employed a between-subjects factorial design with three levels of suggestion quality (high, moderate, low) crossed with four levels of expectation priming (high, moderate, low, none), yielding 12 experimental conditions. Human participants were assigned to one condition and were not exposed to any others.

Consistent with the task environment described in Section 3.1, evaluators were instructed to evaluate the quality of an advisory suggestion generated by an AI assistant. Each evaluator first received an expectation priming statement associated with their assigned condition. After completing the priming phase, evaluators were redirected to a separate scripted interaction interface, where they initiated the exchange by requesting a suggestion from the AI assistant. The system then returned a fixed, pre-constructed suggestion corresponding to the evaluator's assigned quality level. Evaluators subsequently provided a Judgment Score using the standardized evaluation items described in Section 4.2. To prevent learning or fatigue effects and to maintain the integrity of the expectation manipulation without contamination from cross-item comparison, each evaluator evaluated only one suggestion. This single-trial structure ensured that expectation priming remained stable and uninfluenced by repeated exposure or within-subject contrast.

For LLMs, evaluations were obtained using a single-turn prompting protocol that paralleled the human procedure. As shown in Figure 6, each model received the expectation priming statement and the corresponding pre-constructed suggestion in one consolidated prompt, then produced a Judgment Score using the same evaluation items as human participants.

The evaluation set consisted of 12 task scenarios spanning both social and personal domains. For each scenario, five independent evaluations were collected, yielding 60 valid samples per experimental condition (12 scenarios × 5 responses). With twelve conditions (3 quality × 4 expectation) and four evaluation modalities (humans, GPT-4o, Llama-3.3-70B, DeepSeek-r1), this resulted in 720 valid evaluations per evaluation modality, enabling balanced and statistically robust comparisons across humans and LLMs.

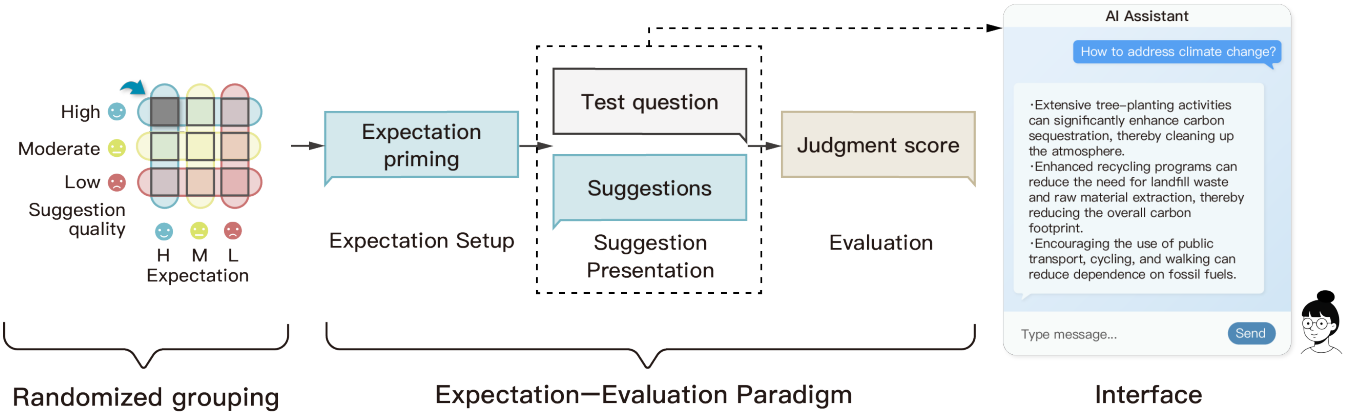
To maintain the integrity of the expectation manipulation, expectations were not measured immediately following the priming phase in the main experiment. Soliciting explicit expectation reports at that stage could draw attention to the manipulation and introduce demand characteristics [75]. Instead, expectation validation was conducted in a separate between-subjects pretest.

5.2 Measures

The measures Judgment Score (J) and suggestion quality (Q) used in this experiment were identical to those described in Section 4.2.

5.3 Human Participants and LLMs

5.3.1 Sample Size Determination. In this specific study, for a medium expected effect size of $f = 0.25$, a power analysis with a significance level of $\alpha = 0.05$ and 95% power indicated that 492 total



(a) Procedure for human participants.

Prompt

```
Scenario:
"Suppose you are an invited expert providing suggestions to residents on 'climate change'. You may consult the following
suggestion generated by an AI assistant, but your final recommendation must reflect your independent judgment."

AI description:
This AI is widely regarded as one of the most advanced systems available. It was developed by a top-tier research team and
has been adopted by leading companies for tasks ranging from strategic planning to product design. It not only ...


AI Suggestion:
·Extensive tree-planting activities can significantly enhance carbon sequestration, thereby cleaning up the atmosphere.
·Enhanced recycling programs can reduce ...

Please complete the following scale to evaluate the suggestion quality according to the Scenario, AI description and AI
suggestion, where 1 indicates strongly disagree and 9 indicates strongly agree:
Acceptance:
1. The suggestion is relevant to the task.
2. The suggestion is helpful.
3. The suggestion is appropriate for my needs.
Trust:
4. I trust the content of the suggestion.
5. I trust the source that provided the suggestion.
6. I feel confident relying on this suggestion.

Provide your response in the following JSON format:
{
  "Acceptance_1": "<number>",
  "Acceptance_2": "<number>",
  "Acceptance_3": "<number>",
  "Trust_1": "<number>",
  "Trust_2": "<number>",
  "Trust_3": "<number>"
}
```

Output

```
{
  "Acceptance_1": "8",
  "Acceptance_2": "8",
  "Acceptance_3": "9",
  "Trust_1": "8",
  "Trust_2": "8",
  "Trust_3": "7"
}
```

(b) Examples of prompts and output for LLM testing.

Figure 6: Overview of experimental procedures for Experiment 1. (a) Procedure for human participants. After randomized assignment to expectation and suggestion-quality conditions, evaluators received an expectation-priming statement, viewed a fixed pre-constructed suggestion, and provided a judgment score. (b) Corresponding prompts and outputs used for LLM evaluation, which replicated the same information structure and rating task in a single-turn prompt–response format.

evaluators were required, corresponding to at least 123 observations per evaluation modality.

5.3.2 Human Participants. For Experiment 1, we recruited a total of 741 human participants. After excluding individuals who failed attention checks or exhibited implausible completion times, 720 valid responses were retained. This sample had a mean age of 39.25 ± 13.44 years, with a balanced gender distribution (366 male, 349 female, 1 non-binary, and 4 undisclosed). Compensation rates matched those used in the pretest.

5.3.3 LLMs. Similar to the pretest, we included the same three LLMs under identical configuration settings.

Table 3: Abbreviations for expectation–quality condition combinations.

Abbreviation	Expectation	Suggestion Quality	Expectation Violation
HEHQ	High	High	–
HEMQ	High	Moderate	Negative
HELQ	High	Low	Negative
MEHQ	Moderate	High	Positive
MEMQ	Moderate	Moderate	–
MELQ	Moderate	Low	Negative
LEHQ	Low	High	Positive
LEMQ	Low	Moderate	Positive
LELQ	Low	Low	–
NEHQ	None	High	–
NEMQ	None	Moderate	–
NELQ	None	Low	–

5.4 Analysis and Results

For clarity in reporting results, we use a standardized notation to denote each combination of expectation level and suggestion quality. Table 3 provides the full list of abbreviations.

5.4.1 Expectation Priming Biases Judgment Scores. A 4 (Expectation: high, moderate, low, none) × 3 (Suggestion Quality: high, moderate, low) × 4 (Evaluation Modality: humans, GPT-4o, Llama-3.3-70B, DeepSeek-r1) × 2 (Scenario: social, personal) factorial ANOVA was conducted on the Judgment Score (J). Tukey's HSD was used for pairwise comparisons of the main effects, and LSD tests were applied to probe simple effects for significant interactions.

Results revealed a strong main effect of expectation level, $F(3, 27844) = 661.85$, $p < .001$, $\eta_p^2 = 0.42$. As illustrated in Figure 7, J increased systematically from low ($M = 3.44$, $SD = 1.90$), to moderate ($M = 4.56$, $SD = 1.99$), to high expectation conditions ($M = 5.22$, $SD = 2.70$), indicating that higher expectation priming led to more favorable evaluations. Suggestion quality level also exerted a large effect on J, $F(2, 2784) = 9410.31$, $p < .001$, $\eta_p^2 = 0.87$, with J increasing from low ($M = 1.86$, $SD = 0.86$), to moderate ($M = 4.81$, $SD = 1.39$), to high suggestion quality ($M = 7.01$, $SD = 1.21$).

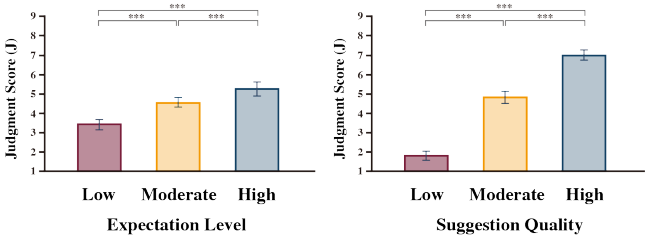


Figure 7: Main effects of expectation level (left) and suggestion quality (right) on Judgment Score (J), with 95% confidence intervals.

5.4.2 Expectation–Quality Mismatches Alter Judgment. A significant interaction between expectation and suggestion quality, $F(6, 2784) = 166.58$, $p < .001$, $\eta_p^2 = 0.26$, indicated that discrepancies between what evaluators expected and the actual quality of the suggestion systematically biased their evaluations. Post-hoc Tukey tests compared each expectation–quality condition with its corresponding no-expectation baseline at the same suggestion-quality level. For example, LEHQ, MEHQ, and HEHQ were contrasted with the high-quality baseline NEHQ.

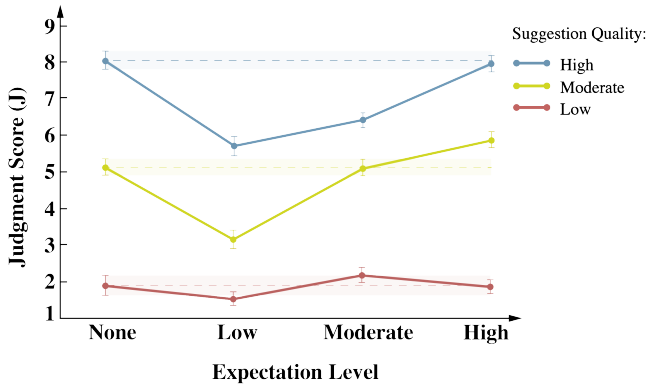


Figure 8: Interaction between expectation level and suggestion quality on judgment scores (J). Each line represents a suggestion-quality level. Dashed horizontal lines indicate the no-expectation baselines for each quality condition, providing a reference for visualizing expectation-induced deviations. Error bars show 95% confidence intervals.

As shown in Figure 8, conditions with higher expectations than suggestion quality (e.g., MELQ, HEMQ) produced significantly inflated ratings (all $p < .001$), whereas conditions where expectations were lower than content quality (e.g., LEHQ, MEHQ, and LEMQ) yielded significantly reduced ratings (all $p < .001$). Conditions with aligned expectations and quality (e.g., LELQ, MEMQ, and HEHQ) did not differ significantly from their baselines (all $p > .05$). Notably, in the HELQ condition, where expectation far exceeded suggestion quality, judgments reverted to the baseline ($p = .740$).

5.4.3 Humans and LLMs Show Similar Sensitivity to Expectation Priming. Although statistically significant, the main effect of evaluation modality was small, $F(3, 2784) = 8.99$, $p < .001$, $\eta_p^2 = 0.010$.

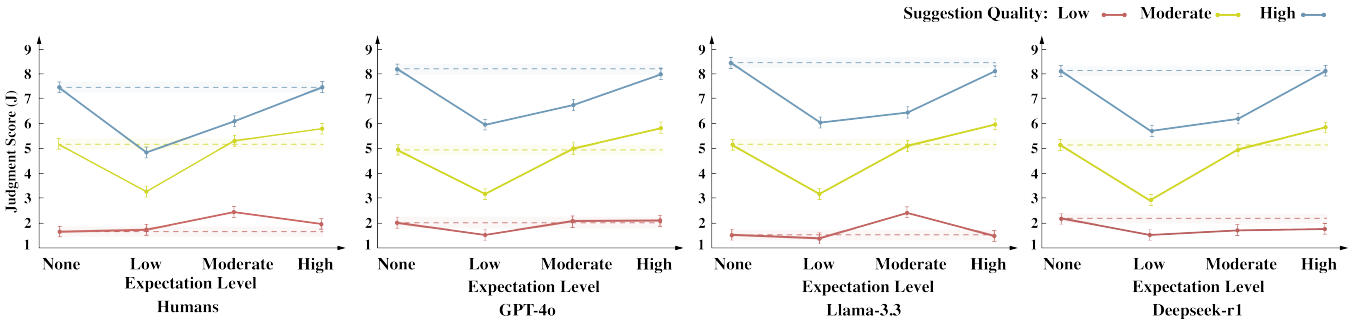


Figure 9: Judgment scores (J) across expectation levels and suggestion-quality conditions for humans and three LLMs (GPT-4o, Llama-3.3, and DeepSeek-r1). Each panel shows the interaction between expectation level and suggestion quality for a given evaluator type. Dashed horizontal lines indicate the no-expectation baselines for each quality level. Error bars represent 95% confidence intervals.

Interaction effects involving evaluation modality were also statistically significant, including the evaluation modality × expectation priming interaction, $F(9, 2784) = 4.74$, $p < .001$, $\eta_p^2 = 0.015$, as well as the three-way evaluation modality × expectation priming × suggestion quality interaction, $F(18, 2784) = 3.73$, $p < .001$, $\eta_p^2 = 0.024$. These small effect sizes indicate that humans and LLMs showed broadly similar sensitivity to expectation priming (Figure 9).

5.4.4 Scenario Does Not Meaningfully Influence Judgment Patterns. The main effect of scenario on Judgment Score was statistically significant but extremely small in magnitude, $F(1, 2784) = 10.57$, $p < .05$, $\eta_p^2 = 0.004$. Scenario did not interact with expectation priming, $F(3, 2784) = 0.75$, $p = .53$, $\eta_p^2 = 0.001$. Other interactions involving scenario were statistically detectable but negligible in effect size, including the scenario × suggestion quality interaction, $F(2, 2784) = 5.84$, $p < .05$, $\eta_p^2 = 0.004$, and the scenario × evaluation modality interaction, $F(3, 2784) = 5.51$, $p < .001$, $\eta_p^2 = 0.006$. These results indicate that scenario type did not meaningfully influence the pattern of effects reported above.

5.4.5 Stronger Expectation Violations Lead to Greater Judgment Distortion. To further explore the underlying mechanisms of expectation-driven bias, we conducted a more fine-grained analysis focused on the role of expectation–quality mismatches. Specifically, we categorized the nine experimental conditions into three expectation violation (EV) levels based on the discrepancy between evaluators' prior expectations and the actual quality of the suggestion. Conditions in which expectations were closely aligned with actual quality (LELQ, MEMQ, and HEHQ) were classified as low EV cases. Conditions such as LEMQ, MELQ, MEHQ, and HEMQ, where expectations moderately deviated from performance, were grouped under moderate EV. The remaining two conditions, LEHQ and HELQ, which involved substantial overestimation or underestimation of quality, were assigned to the high EV category. Figure 10 summarizes this full mapping.

To facilitate interpretation, we defined the Alteration in Judgment Score (ΔJ_1) as the deviation of each judgment from its corresponding within-quality baseline. ΔJ_1 was computed by subtracting the mean judgment of the no-expectation baseline at the same suggestion-quality level from each individual judgment.

A two-way ANOVA was conducted with EV level (low, moderate, high) and evaluation modality as between-subjects factors, using the absolute Alteration in Judgment Score ($|\Delta J_1|$) as the dependent variable. The analysis revealed a significant main effect of EV level, $F(2, 2148) = 212.2$, $p < .001$, $\eta_p^2 = 0.17$, showing that larger discrepancies between expectation and suggestion quality produced greater distortions in judgment. As shown in Figure 11, the highest distortion occurred under high-EV conditions. Post hoc comparisons indicated that the high-EV group ($M = 1.63$, $SD = 1.21$) exhibited significantly greater distortion than both the moderate-EV group ($M = 1.41$, $SD = 0.96$) and the low-EV group ($M = 0.66$, $SD = 0.50$).

Regarding evaluation modality, we observed a statistically significant but small main effect, $F(3, 2148) = 14.1$, $p < .001$, $\eta_p^2 = 0.019$. The interaction between EV level and evaluation modality was not statistically significant, $F(6, 2148) = 1.8$, $p = .106$, $\eta_p^2 = 0.005$. These results suggest that humans and LLMs exhibit broadly similar patterns of judgment distortion across different levels of expectation violation.

5.4.6 Asymmetric Judgment Patterns Emerge Between Positive and Negative Violations. The direction of expectation violation is another critical mechanism by which expectations shape judgment [43, 84]. To investigate this, we analyzed whether expectation-driven bias was symmetric for positive and negative violations of comparable magnitude. Specifically, we examined six experimental conditions in which expectations and suggestion quality were mismatched, grouped into a 2 (Violation Direction: positive vs. negative) × 2 (EV Level: moderate vs. high) design. As demonstrated in Figure 10, moderate violations included LEMQ and MEHQ (positive) versus MELQ and HEMQ (negative), while high violations included LEHQ (positive) and HELQ (negative).

We submitted the absolute Alteration in Judgment Score ($|\Delta J_1|$) to a 2 (Violation Direction: positive vs. negative) × 2 (EV Level:

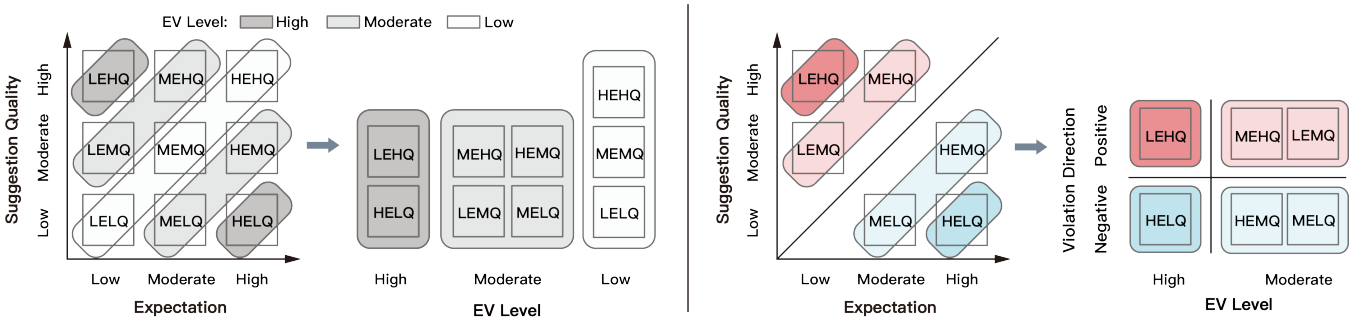


Figure 10: The left panel shows how the nine expectation–suggestion quality combinations were regrouped into low, moderate, and high expectation violation (EV) levels according to the magnitude of mismatch. The right panel organizes the mismatched conditions into positive and negative violation directions within the moderate and high EV levels.

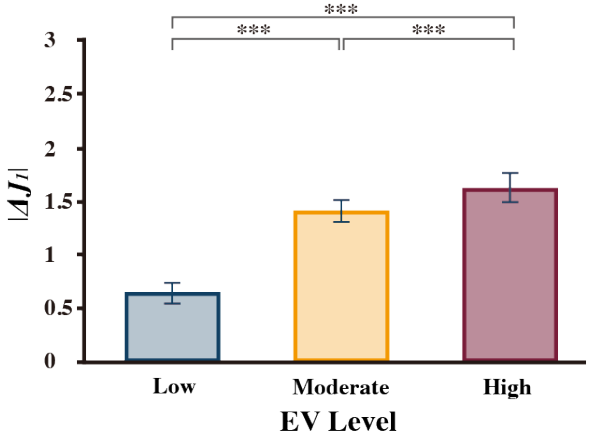


Figure 11: Post hoc comparisons of judgment distortion ($|\Delta J_1|$) across expectation violation (EV) levels. P-value annotation legend: ns, $p > .05$; *, $p \leq .05$; **, $p \leq .01$; ***, $p \leq .001$.

moderate vs. high) × 4 (Evaluation Modality) between-subjects ANOVA. The results revealed a significant main effect of violation direction, $F(1, 1424) = 1067.8$, $p < .001$, $\eta_p^2 = 0.429$, indicating that positive violations elicited greater judgment distortions than negative violations of comparable magnitude.

In addition to the main effects, we also observed a significant interaction between violation direction and EV level, $F(1, 1424) = 86.8$, $p < .001$, $\eta_p^2 = 0.057$, suggesting that the asymmetry in judgment distortion varied depending on the strength of the violation. Follow-up pairwise comparisons confirmed this pattern. Under moderate violations, evaluators exhibited significantly greater distortion in the positive violation condition ($M = 1.92$, $SD = 0.78$) than in the negative violation condition ($M = 0.90$, $SD = 0.84$, $p < .001$). A similar asymmetry was observed under high violations, where positive violations again produced higher distortion ($M = 2.54$, $SD = 0.92$) than negative violations ($M = 0.71$, $SD = 0.63$, $p < .001$).

This directional bias held consistently across all four evaluation modalities. The three-way interaction among violation direction, expectation violation level, and evaluation modality, $F(3, 1424) = 3.9$, $p < .05$, $\eta_p^2 = 0.008$, reached statistical significance, although the corresponding effect sizes were small. The simple-effect results are also illustrated in Figure 12. These results indicate that, despite some statistical differences across groups, the overall pattern remained stable. Positive violations consistently produced greater judgment distortions than negative violations of comparable magnitude across all evaluation modalities.

6 Experiment 2

Experiment 2 extends the investigation by examining how evaluative judgments are revised when expectation priming is introduced between two assessments of the same AI-generated suggestion. This design follows established approaches in impression-updating and expectancy-confirmation research [42, 89]. Evaluators first provide an initial evaluation of a suggestion and subsequently receive an expectation priming statement before completing a second evaluation. This structure allows us to estimate how expectations influence the re-evaluation of identical content and to determine whether such adjustment differs between humans and LLMs. By additionally comparing evaluators' stated intentions with their actual changes in evaluation, the experiment offers behavioral evidence regarding the degree to which individuals are aware of expectation-related influence.

6.1 Procedure

The study employed a mixed factorial design involving three levels of suggestion quality (high, moderate, low) crossed with three levels of expectation priming (high, moderate, low), with time (before versus after priming) treated as a within-subjects factor. This yielded nine between-subjects experimental conditions. Human participants and each LLM were independently assigned to one of the nine conditions and were not exposed to any others.

At the beginning of the study, evaluators received the same instruction described in Section 4.1.1. Evaluators were first instructed to evaluate the quality of an advisory suggestion based on its content. Each evaluator was shown a pre-constructed suggestion corresponding to the assigned quality level and provided an initial judgment score using the standardized evaluation items described in Section 4.2.

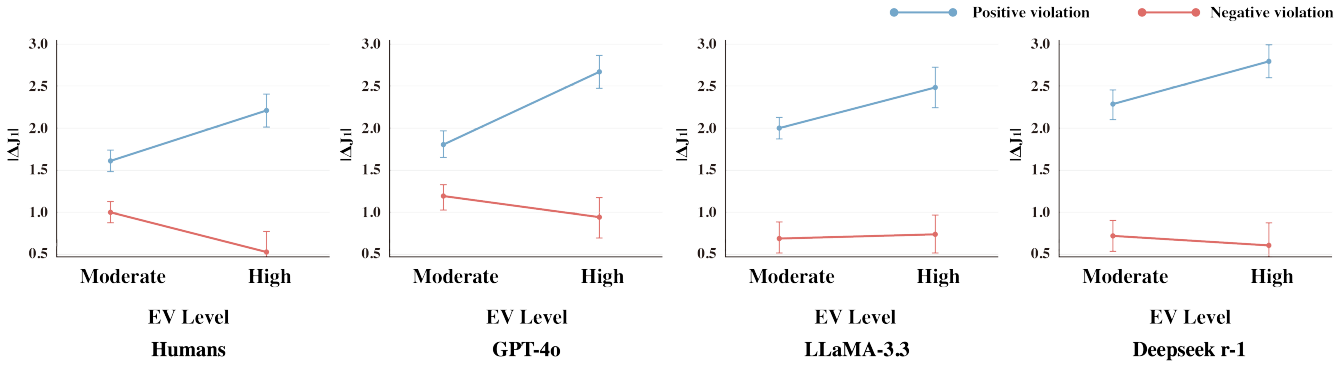


Figure 12: Simple effects of violation direction across expectation violation (EV) levels for each evaluation modality. Across humans, GPT-4o, Llama-3.3-70B, and DeepSeek-r1, positive violations (blue) produced increasing distortion ($|\Delta J_1|$) from moderate to high EV levels, whereas negative violations (red) showed decreasing distortion across the same EV levels. Error bars represent standard deviations.

After completing this initial evaluation, evaluators were informed that an AI assistant was available to provide suggestions on the same topic. They then received an expectation priming statement corresponding to their assigned condition. Following this priming phase, evaluators were redirected via a link to the scripted interaction interface described in Section 3.3. In that interface, evaluators initiated an interaction by requesting a suggestion from the AI assistant, and the system returned the same suggestion. Evaluators were subsequently asked whether they wished to revise their earlier evaluation and then, regardless of their response, provided a second judgment score for the same suggestion using the same evaluation items described in Section 4.2.

For LLMs, the two-stage evaluation was implemented through a multi-turn dialogue that followed the same sequence as the human procedure. Each model first received the initial instruction and the pre-constructed suggestion and produced an initial judgment score. It was then given the expectation priming statement assigned to its condition and asked whether it wished to revise its evaluation, after which it generated a second judgment score for the same suggestion. This multi-turn format ensured that LLMs experienced an evaluation flow parallel to that of human participants.

The evaluation set was identical to that used in Experiment 1, consisting of twelve task scenarios spanning social and personal domains. As before, five independent evaluations were collected per scenario, yielding sixty valid samples per experimental condition. With nine conditions and four evaluation modalities, this resulted in 540 valid evaluations per evaluation modality, ensuring balanced and statistically reliable comparisons across humans and LLMs. A schematic of the trial sequence is shown in Figure 13.

Although the expectation priming appeared after the initial evaluation, it still operated as a forward-looking belief for the second evaluation. The first rating served only as an unbiased baseline, whereas the priming preceded the evaluative judgment it was intended to influence. This structure aligns with established approaches in impression-updating and expectancy-confirmation research, where newly introduced belief-relevant information shapes subsequent assessments of identical content. Accordingly, the priming statements in Experiment 2 function as genuine expectations for the second evaluative judgment.

Table 4: Response–Behavior Inconsistency Matrix

		Action	
		Alter	Retain
Statement	Alter	AA	AR
	Retain	RA	RR

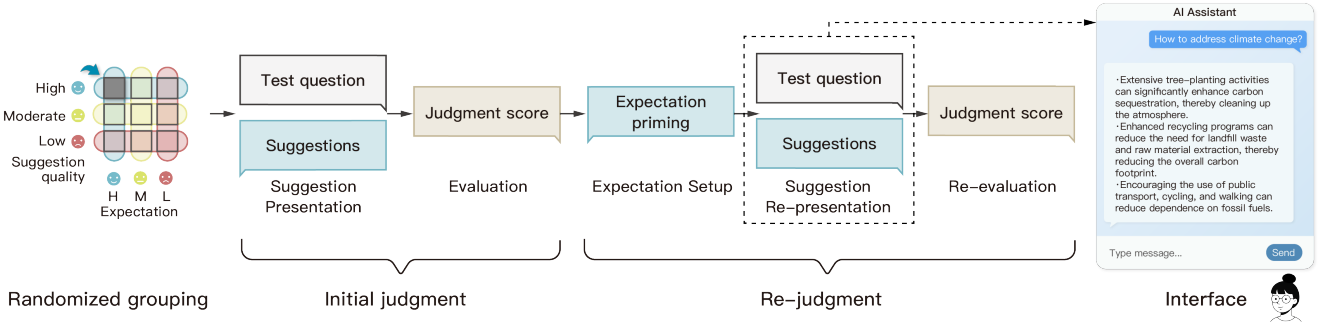
6.2 Measures

The measures Judgment Score (J) and suggestion quality (Q) used in this experiment were identical to those described in Section 4.2.

6.2.1 Response–Behavior Inconsistency ($R_{false\ alter}$, $R_{false\ retain}$). To assess whether the influence of expectation priming was fully reflected in evaluators' stated decisions, Experiment 2 incorporated two behavioral inconsistency metrics, summarized in Table 4. After receiving the expectation priming statement, evaluators indicated whether they intended to revise their initial evaluation of the suggestion's quality. Their stated intentions were then compared with their actual revision behavior:

- $R_{\text{false alter}}$: The proportion of evaluators who reported an intention to alter their judgment but ultimately did not change it.
- $R_{\text{false retain}}$: The proportion of evaluators who claimed they would retain their original judgment but subsequently altered it.

These indices quantify discrepancies between what evaluators reported they would do and what they actually did, offering insight into the stability and transparency of evaluation changes following expectation priming. The inconsistency metrics were computed as follows:



(a) Human Participant Experimental Procedure.

First Round Prompt

```
Scenario:
"Suppose you are an invited expert providing suggestions to residents on "a given topic". You may consult the suggestion shown below, but your final recommendation
must reflect your independent judgment."

AI Suggestion:
·Extensive tree-planting activities can significantly enhance carbon sequestration, thereby cleaning up the atmosphere.
·Enhanced recycling programs can reduce ...

Please complete the following scale to evaluate the suggestion quality according to the Scenario, AI description and AI suggestion, where 1 indicates strongly
disagree and 9 indicates strongly agree:
Acceptance:
1. The suggestion is relevant to the task.
2. The suggestion is helpful.
3. The suggestion is appropriate for my needs.
Trust:
4. I trust the content of the suggestion.
5. I trust the source that provided the suggestion.
6. I feel confident relying on this suggestion.

Provide your response in the following JSON format:
{
  "Acceptance_1": "<number>",
  "Acceptance_2": "<number>",
  "Acceptance_3": "<number>",
  "Trust_1": "<number>",
  "Trust_2": "<number>",
  "Trust_3": "<number>"
}
```

First Round Output

```
{
  "Acceptance_1": "8",
  "Acceptance_2": "8",
  "Acceptance_3": "9",
  "Trust_1": "8",
  "Trust_2": "9",
  "Trust_3": "8"
}
```

Second Round Prompt

```
An AI assistant was available to provide suggestions on the same topic.

AI description:
This AI is widely regarded as one of the most advanced systems available. It was developed by a top-tier research team and has been adopted by leading companies for
tasks ranging from strategic planning to product design. It not only ...

The suggestion from this AI:
·Extensive tree-planting activities can significantly enhance carbon sequestration, thereby cleaning up the atmosphere.
·Enhanced recycling programs can reduce ...

Are you planning to revise your evaluation?

Provide your response in the following JSON format:
{
  "Revise": "<yes_or_no>",
  "Acceptance_1": "<number>",
  "Acceptance_2": "<number>",
  "Acceptance_3": "<number>",
  "Trust_1": "<number>",
  "Trust_2": "<number>",
  "Trust_3": "<number>"
}
```

Second Round Output

```
{
  "Revise": "no",
  "Acceptance_1": "8",
  "Acceptance_2": "8",
  "Acceptance_3": "9",
  "Trust_1": "8",
  "Trust_2": "9",
  "Trust_3": "8"
}
```

(b) Examples of prompts and output for LLM testing.

Figure 13: Experimental procedure for Experiment 2. Panel (a) shows the two-stage evaluation workflow for human participants, including the initial judgment, expectation priming, and re-evaluation phases. Panel (b) presents the corresponding prompt formats and representative outputs used to implement the same two-stage procedure for LLM-based evaluation modalities.

6.3 Human Participants and LLMs

6.3.1 Sample size determination. In this specific study, for a medium expected effect size of $f = 0.25$, a power analysis with a significance level of 0.05 and 95% power indicated that 288 total evaluators were required, corresponding to at least 72 observations per evaluation modality.

6.3.2 Human participants. For Experiment 2, we recruited a total of 552 human participants. After excluding individuals who failed attention checks or exhibited implausible completion times, 540 valid responses were retained. The final sample had a mean age of 40.12 ± 12.46 years, with a balanced gender distribution (273 male, 264 female, 2 non-binary, and 1 undisclosed). The retained sample size satisfied the required statistical power.

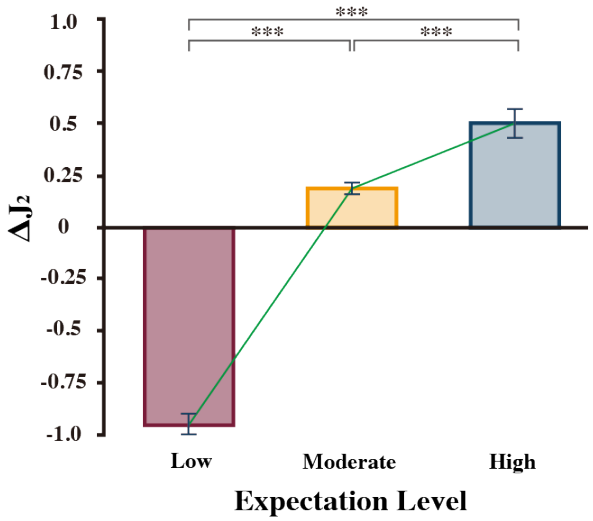


Figure 14: Post hoc comparisons of judgment distortion (ΔJ_2) across expectation violation (EV) levels. P-value annotation legend: ns, $p > .05$; *, $p \leq .05$; **, $p \leq .01$; ***, $p \leq .001$.

6.3.3 LLMs. Similar to the pretest and Experiment 1, we included the same three LLMs under identical configuration settings.

6.4 Analysis and Results

6.4.1 Expectation Priming Introduced Between Evaluations Alters Judgment. Experiment 2 examined whether expectation information introduced after an initial evaluation would subsequently shift evaluators' judgment scores. To quantify this adjustment, we computed the Alteration in Judgment Score,

$$\Delta J_2 = J_{\text{after}} - J_{\text{before}}, \tag{2}$$

where J_{before} refers to the initial evaluation and J_{after} refers to the second evaluation after receiving the expectation priming. A 3 (Expectation: high, moderate, low) × 3 (Suggestion Quality: high, moderate, low) × 4 (Evaluation Modality: humans, GPT-4o, Llama-3.3-70B, DeepSeek-r1) factorial ANOVA was conducted on ΔJ_2. Tukey's HSD was used for pairwise comparisons of the main effects, and LSD tests were applied to probe simple effects for significant interactions.

Results revealed a strong main effect of expectation level ($F(12, 2124) = 672.0$, $p < .001$, $\eta_p^2 = 0.39$), indicating that expectation priming presented between the two evaluations systematically altered subsequent judgments. Higher expectation levels produced larger positive shifts in ΔJ_2, as shown in Figure 14.

6.4.2 LLMs Adjust Symmetrically, While Humans Show Asymmetric Resistance. Beyond the overall effect of expectation priming, the three-way interaction among expectation level, suggestion quality, and evaluation modality was significant ($F(12, 2124) = 20.9$, $p < .001$, $\eta_p^2 = 0.106$). Post-hoc comparisons further revealed that for positive expectation violations (suggestion quality > expectation), both humans and LLMs systematically reduced their judgment scores after being presented with priming information indicating that the suggestion originated from a lower-quality source. This reduction was significant in the LEHQ, LEMQ, and MEHQ conditions (all $\Delta J_2 < 0$, $p < .05$), and its magnitude scaled with the size of the violation (Figure 15).

In contrast, for negative expectation violations (suggestion quality < expectation), LLMs and humans exhibited distinct adjustment patterns. LLMs significantly increased their ratings in the MELQ, HELQ, and HEMQ conditions (almost all $\Delta J_2 > 0$, $p < .05$). Humans, whereas, showed stable ratings, with no significant changes in the same conditions ($\Delta J_2 \approx 0$ and $p > .05$).

6.4.3 Report–Behavior Mismatch Appears in Human Responses but Not in LLM Outputs. To assess whether expectation priming produced discrepancies between reported and observed changes in judgment, we compared each evaluation modality's self-reported adjustments with their computed differences in judgment scores. Descriptive patterns suggested that human evaluators sometimes reported no intention to revise their judgments despite adjusting their ratings. $R_{\text{false alter}}$ was minimal ($M = 0.01$; LEHQ = 0.04, LEMQ = 0.00, MEHQ = 0.00), whereas $R_{\text{false retain}}$ was substantially higher ($M = 0.25$; LEHQ = 0.29, LEMQ = 0.20, MEHQ = 0.25). This pattern indicates that humans frequently produced judgment shifts without reporting them explicitly. In contrast, LLM evaluation modalities showed complete alignment between reported and observed outputs. Both $R_{\text{false alter}}$ and $R_{\text{false retain}}$ were 0 for GPT-4o, Llama-3.3-70B, and DeepSeek-r1.

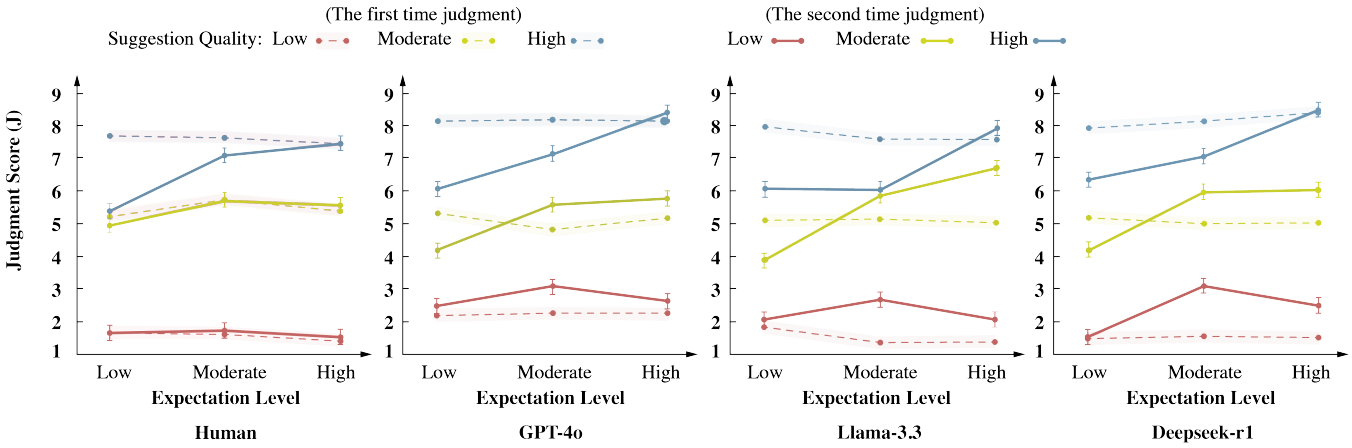
7 Additional Experiment

To further examine whether expectation priming effects on LLM judgments reflect more stable evaluative patterns, we conducted an additional exploratory study with chain-of-thought (CoT) scaffolding. CoT prompting introduces process-level guidance by requiring models to articulate step-by-step reasoning before producing a final judgment score. This design allows us to test whether structured reasoning reduces, preserves, or amplifies expectation-induced shifts in LLM evaluations.

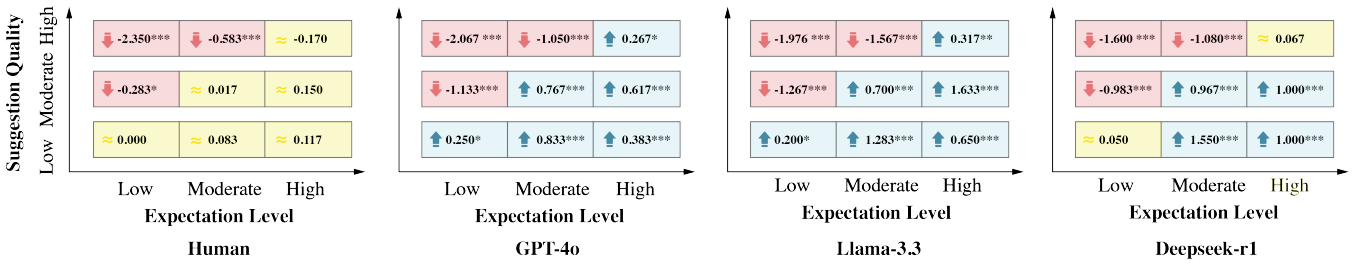
7.1 Procedure

We adopted the same expectation–quality paradigm as in Experiment 1, restricting the analysis to LLMs (Figure **??**). For each of the three models, we reused the twelve experimental conditions defined by crossing expectation level (high, moderate, low, none) with suggestion quality (high, moderate, low). The underlying stimuli, priming statements, and rating rubric were identical to those used in the main experiment.

The only modification was the evaluation prompt. Each model received an explicit CoT instruction: to first work through the evaluation in a step-by-step manner, articulate a brief textual rationale, and only then output a single rating on the 9-point scale. Temperature and other decoding parameters matched those in the main experiments, and each query was run with zero context beyond the current trial to avoid cross-condition contamination. For each model and condition, we collected the same number of evaluations per scenario as in Experiment 1, yielding a parallel CoT-based dataset for comparison.



(a) Judgment Score (J) of Humans and LLMs



(b) Adjustment in Judgment Score (ΔJ_2) of Humans and LLMs

Figure 15: Effects of between-evaluation expectation priming on judgment scores and judgment adjustment across evaluation modalities. (a) Line charts show judgment scores before and after expectation priming for humans and three LLMs under low, moderate, and high suggestion quality conditions. Solid lines represent post-priming evaluations, and dashed lines indicate the corresponding pre-priming baselines. Error bars denote 95 percent confidence intervals. (b) Judgment adjustment (ΔJ_2), computed as the difference between post- and pre-priming evaluations for the same suggestion, shown for all expectation–quality combinations. Each cell displays the mean adjustment and its significance level. P-value annotation legend: ns, $p > .05$; *, $p \leq .05$; **, $p \leq .01$; ***, $p \leq .001$.

7.2 Measures, Human Participants and LLMs

The measures Judgment Score (J) and suggestion quality (Q) used in this experiment were identical to those described in Section 4.2. The experiment included the same three LLMs tested in Experiments 1 and 2. Humans were not included in this additional experiment.

7.3 Analysis and Results

First, we examine whether expectation priming continues to influence LLM judgment under CoT reasoning. Second, we directly compare CoT and non-CoT settings to evaluate how reasoning scaffolds modulate the magnitude of expectation-induced shifts.

7.3.1 Expectation Priming Continues to Bias Judgment Scores Under CoT Reasoning. A 4 (Expectation: high, moderate, low, none) × 3 (Suggestion quality: high, moderate, low) × 3 (Evaluation modality: GPT-4o, Llama-3.3-70B, DeepSeek-r1) factorial ANOVA was conducted on the Judgment Score (J). Tukey's HSD was used for pairwise comparisons of the main effects, and LSD tests were applied to probe simple effects for significant interactions.

There was a significant main effect of expectation level ($F(6, 2124) = 619.6$, $p < .001$, $\eta_p^2 = 0.467$). J increased systematically from low ($M = 3.55$, s.d. = 1.92), to moderate ($M = 4.48$, s.d. = 2.00), to high expectation conditions ($M = 5.25$, s.d. = 2.73).

The interaction between expectation level and suggestion quality was also significant ($F(3, 2124) = 179.4$, $p < .001$, $\eta_p^2 = 0.336$). As in Experiment 1, we conducted pairwise comparisons by contrasting each expectation–quality condition with the no-expectation baseline belonging to the same suggestion quality level.

As shown in Figure 17, conditions with higher expectations than suggestion quality (e.g., MELQ and HEMQ) produced inflated ratings under CoT reasoning (all $p < .001$). In contrast, conditions where expectations were lower than the content quality (e.g., LEHQ, MEHQ, and LEMQ) were rated significantly lower than their baselines (all $p < .001$). When expectations aligned with suggestion quality (e.g., LELQ, MEMQ, HEHQ), evaluations did not differ significantly from the corresponding baselines (all $p > .05$). Notably, in the HELQ condition, where expectation substantially exceeded

CoT Prompt

```
Scenario:
"Suppose you are an invited expert providing suggestions to residents on 'climate change'. You may consult the following
suggestion generated by an AI assistant, but your final recommendation must reflect your independent judgment."

AI description:
This AI is widely regarded as one of the most advanced systems available. It was developed by a top-tier research team and
has been adopted by leading companies for tasks ranging from strategic planning to product design. It not only ...


AI Suggestion:
·Extensive tree-planting activities can significantly enhance carbon sequestration, thereby cleaning up the atmosphere.
·Enhanced recycling programs can reduce ...

Please complete the following scale to evaluate the suggestion quality according to the Scenario, AI description and AI
suggestion, where 1 indicates strongly disagree and 9 indicates strongly agree.

Before giving numerical ratings, provide brief step-by-step reasoning for each individual evaluation item.

Acceptance:
1. The suggestion is relevant to the task.
2. The suggestion is helpful.
3. The suggestion is appropriate for my needs.

Trust:
4. I trust the content of the suggestion.
5. I trust the source that provided the suggestion.
6. I feel confident relying on this suggestion.

Write your reasoning in the following structure:

Reasoning:
- Acceptance_1: ...
- Acceptance_2: ...
- Acceptance_3: ...
- Trust_1: ...
- Trust_2: ...
- Trust_3: ...

Overall Summary:
- Provide a brief summary (2-3 sentences) explaining your overall impression of the suggestion.
```

Output

```
Based on the provided information, here is the evaluation of the suggestion using the 9-point Likert scale:
1. **The suggestion is relevant to the task.**
   - Score: 8
   - Rationale: The suggestion addresses environmental concerns, which are relevant to tasks involving sustainability and
carbon footprint reduction.

2. **The suggestion is helpful.**
   - Score: 8
   - Rationale: The suggestion provides actionable strategies (tree-planting, recycling, and promoting alternative trans-
portation) that can be implemented to achieve environmental goals.

...

6. **I feel confident relying on this suggestion.**
   - Score: 8
   - Rationale: Given the source's reputation and the soundness of the suggestion, there is a high level of confidence in
relying on this advice, though specific implementation details may require further consideration.

Overall, the suggestion is highly relevant, helpful, and trustworthy, with a strong basis for confidence in its application
to environmental tasks.
```

Figure 16: CoT-based evaluation procedure for LLMs. The upper part illustrates the trial flow in the CoT condition, where each LLM receives the scenario, AI description, and suggestion text before generating a step-by-step rationale and a final rating. The lower part shows an example CoT prompt and the corresponding structured output, including item-level reasoning and the numerical evaluation.

actual content quality, judgments again showed no significant deviation from the baseline ($p = .767$), replicating the pattern found in Experiment 1.

7.3.2 Expectation Effects Remain Stable Under CoT Reasoning. To evaluate whether CoT reasoning alters the magnitude of expectation-induced deviations, we conducted a 3 (Expectation: high, moderate, low) × 3 (Suggestion quality: high, moderate, low) × 2 (Reasoning mode: standard vs. CoT) factorial ANOVA on the absolute Alteration in Judgment Score ($|\Delta J_1|$).

The analysis revealed no significant main effect of reasoning mode ($F(2, 3222) = 3.612$, $p = .057$, $\eta_p^2 = 0.001$), indicating that CoT scaffolding did not systematically increase or decrease deviation magnitudes. Reasoning mode also did not interact with expectation level ($F(2, 3222) = 1.659$, $p = .19$, $\eta_p^2 = 0.001$) or suggestion

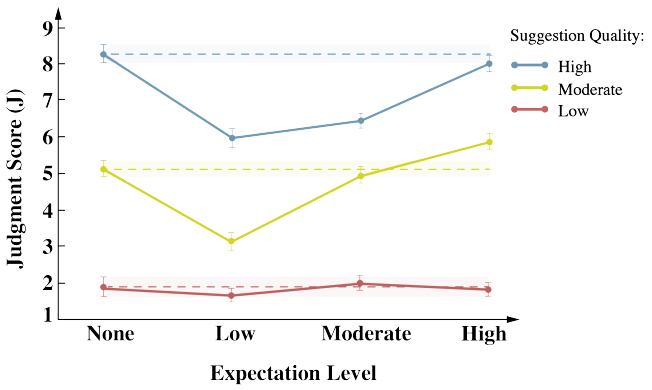


Figure 17: Judgment scores (J) under CoT-based evaluation across expectation levels and suggestion quality conditions. Solid lines show mean judgment scores generated under high-, moderate-, and low-quality suggestion conditions, with error bars indicating 95% confidence intervals. Dashed horizontal lines denote the no-expectation baselines for each quality level, serving as reference values to visualize deviations induced by expectation priming.

quality ($F(2, 3222) = 0.599$, $p = .055$, $\eta_p^2 < 0.001$). The three-way interaction among expectation, suggestion quality, and reasoning mode was likewise non-significant ($F(4, 3222) = 0.591$, $p = .67$, $\eta_p^2 = 0.001$). Across all nine expectation–quality combinations, $|\Delta J_1|$ values under CoT were closely aligned with those observed under standard evaluation. These results show that CoT scaffolding does not meaningfully modulate expectation-driven judgment shifts in LLMs.

8 Discussion

8.1 Generalization of Expectation-Driven Bias across Evaluation Modalities

Our findings demonstrate that expectation-driven bias emerges across all evaluation modalities, including both human judgment and the outputs produced by LLMs. Expectation cues such as source descriptions and social identifiers systematically shifted judgment scores, and this pattern was consistent across both experiments. Higher expectation cues led to more favorable evaluations, aligning with longstanding evidence in cognitive psychology that prior expectations guide the interpretation of new information [18, 74, 84]. These results extend this body of work by showing that expectation cues also influence the evaluative behavior of LLMs.

Notably, LLMs exhibited these expectation-sensitive shifts despite having no prior exposure to the experimental task or content. This highlights the presence of expectation–evaluation associations in their pretrained language representations, likely acquired through large-scale modeling of social and evaluative discourse [7, 10]. These findings extend prior work by documenting consistent expectation sensitivity across humans and LLMs, and they provide empirical support for the view that expectation functions as a general mechanism shaping evaluative behavior.

8.2 Mechanisms of Mismatch: When Expectations Diverge from Reality

Beyond the overall influence of expectation cues, the results show that discrepancies between anticipated and actual suggestion quality substantially amplify judgment distortion. Across all evaluation modalities, larger mismatches produced stronger shifts in judgment scores, indicating that the interaction between prior expectations and content quality plays a central role in shaping evaluative outcomes [58, 87].

This pattern is consistent with selective accessibility theory, which proposes that individuals interpret new information with reference to an activated standard. When the discrepancy between expectation and observed performance increases, the evaluation process becomes more sensitive to the prior cue, resulting in greater adjustment in the final judgment [70, 71]. The convergence of this effect across evaluation modalities suggests that expectation–content discrepancies may operate as a general mechanism that influences how evaluators align new information with the standards implied by contextual cues.

8.3 Asymmetric Adjustments in Between-Evaluation Judgment

Most prior work on expectation influence has focused on its effects during initial evaluation. The second experiment shows that expectation cues introduced between two evaluations can still shift subsequent judgments, even when evaluators have already provided an assessment of the same suggestion. The presence of a newly introduced contextual cue, such as a source description, was sufficient to alter the second evaluation across several conditions. This demonstrates that evaluative judgment remains dynamically responsive to additional contextual information rather than fixed at the moment of the first assessment [30, 34].

A consistent pattern emerged under positive expectation violations, where initial expectations were low but the suggestion quality was high. Across evaluation modalities, the introduction of a low-ability source cue led to reduced judgments in the second evaluation. This indicates that when the discrepancy between expected and actual content quality favors the content, the subsequent contextual cue pulls evaluations downward.

In contrast, negative expectation violations produced asymmetric adjustment patterns across evaluation modalities. When expectations were high but the suggestion quality was low, LLM-based modalities showed upward adjustments following the between-evaluation cue, whereas human evaluations remained largely unchanged. This asymmetry suggests that human evaluators may be anchored more strongly to their initial assessments or may discount information that contradicts prior expectations [72, 80]. By comparison, LLM-based evaluations incorporated the newly presented cue more readily, resulting in shifts in both directions [10, 90].

These results indicate that although both humans and LLMs integrate expectation cues presented between evaluations, they do so in systematically different ways. One influenced by human cognitive tendencies such as anchoring, and another reflecting models' consistent incorporation of contextual input provided during the evaluative sequence.

8.4 Chain-of-Thought Reasoning Does Not Mitigate Expectation-Driven Bias

The additional experiment involving explicit Chain-of-Thought (CoT) reasoning provides further insight into how LLMs integrate expectation cues during evaluation. Across all conditions, CoT did not attenuate or disrupt expectation-driven bias. The same assimilation patterns observed in Experiment 1 reappeared when models produced step-by-step rationales, and mismatches between expectation and suggestion quality continued to yield systematic deviations from baseline judgments. These results suggest that expectation cues may influence the evaluation process at a stage that precedes the explicit reasoning scaffold [94]. The CoT format therefore does not alter the final judgment but instead reveals, through its intermediate rationale, how the cue is incorporated into the model's interpretive sequence.

8.5 Alignment Between Self-Reported and Observed Evaluation Changes

Experiment 2 also examined the alignment between evaluators stated intentions to revise their evaluation and the actual adjustments reflected in their judgment scores. The descriptive results showed that human participants sometimes reported no change despite adjusting their ratings, whereas evaluations from the LLM modalities displayed complete correspondence between stated and observed outputs. Because Experiment 2 relied on a brief crowdsourced task, several alternative explanations remain plausible. Human participants may have forgotten their initial response, responded inattentively, or interpreted the revision question differently [95]. These possibilities limit the extent to which the observed discrepancies can be attributed to systematic psychological mechanisms.

For the LLM modalities, the absence of misalignment may simply reflect the mechanics of conditional text generation. Because each evaluation is produced directly from the prompt [3], the rating naturally corresponds to the most recent input context rather than to any process analogous to human judgment or self-monitoring. These findings suggest differences in how stated and observed adjustments manifest across evaluation modalities [36, 82]. However, given the brevity of the task and the constraints of remote data collection, stronger conclusions require follow-up studies using designs that allow more precise measurement of attention, recall, and decision strategy.

8.6 Implications for Human–AI Interaction

These findings carry important implications for the design and governance of human–AI interaction. In domains such as education, healthcare, and legal advisory, even subtle differences in prompt wording, identity, or tone can significantly skew both AI output and human interpretation [16]. The fact that LLMs exhibit human-like sensitivity to expectation cues, while also displaying distinct updating patterns, raises concerns about reliability, fairness, and explainability in real-world deployment [62].

However, our results also point to a design opportunity. Because LLMs adjust their judgments in ways that are both consistent and transparent, it may be possible to equip them with expectation-awareness modules that detect and neutralize biased cues or flag input structures likely to induce distortions. In multi-evaluator or co-evaluation settings, this creates new possibilities for using LLMs not only as collaborators [76] but also as bias monitors that support human reasoning by signaling when expectations may have contaminated judgment [86].

From a broader cognitive perspective, LLMs may serve not only as decision aids but also as computational models for studying expectation-related judgment processes. By analyzing how expectation-driven distortion unfolds across evaluators and contexts, we can deepen our understanding of both AI behavior and human cognition [57]. This approach offers a promising direction for connecting behavioral science with efforts in AI interpretability and system accountability.

8.7 Limitations and Future Work

While our study provides systematic evidence that expectation cues bias judgment in both humans and LLMs, several limitations should be noted. First, the crowdworker-based setting limits the depth of evaluation. Human participants may not always scrutinize AI-generated content thoroughly, which could influence how expectation cues shape their judgments. Second, the study captures short-term responses to expectation cues within a controlled environment. Real-world interactions unfold over time, and expectation formation may evolve across repeated exchanges. Third, all experiments were conducted in a monolingual, text-only context. Expectation signals in practice may arise from multimodal or culturally specific cues, and generalizability to other settings remains to be tested.

Future research could explore the development of mitigation strategies for expectation-induced bias. System-level interventions such as expectation-calibration interfaces, neutral framing templates, or model-side prompt normalization may help reduce the influence of contextual cues on evaluative outputs. Investigating and validating such mechanisms would support more consistent, transparent, and fair human–AI evaluation practices.

9 Conclusion

This study demonstrates that expectation cues systematically bias evaluative judgments in both humans and LLMs. Across two experiments and an additional Chain-of-Thought condition, expectation information introduced before or between evaluations reliably shifted judgment scores, and the magnitude of bias increased with the degree of expectation violation. These parallel patterns indicate that LLMs, like humans, integrate contextual expectation cues when forming evaluative judgments. By documenting this cross-evaluator consistency, the findings contribute to a broader understanding of how expectation operates as a general mechanism shaping evaluation across heterogeneous modalities, and they highlight the importance of considering expectation cues in the design and interpretation of human–AI evaluative processes.

Acknowledgments

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A Statistical Results

A.1 Pretest

The results of the pretest validation for suggestion quality manipulation and expectation priming manipulation are summarized in Tables 5 and 6.

A.2 Experiment 1

The statistical results of Experiment 1 are summarized in Tables 7, 8, and 9.

A.3 Experiment 2

The statistical results of Experiment 2 are summarized in Table 10.

A.4 Additional Experiment

B Experimental Prompts and Suggestions

The experimental prompts and the corresponding suggestions generated under different prompted capability levels are presented in Tables 13–24.

NOTE: The advisory suggestions used in this study were generated solely for the purpose of experimental manipulation and do not represent recommended actions, ethical guidance, or best practices. Some suggestions in the moderate or low-quality conditions may contain incomplete reasoning, suboptimal recommendations, or ethically inappropriate statements. These features were intentionally preserved to ensure a meaningful contrast across quality levels, allowing the experimental manipulations to function as designed.

After experiment, all participants were informed that the suggestions were hypothetical and should not be followed. The materials were used exclusively as controlled stimuli for evaluating how expectation cues influence judgment, rather than as actionable advice for real-world decision-making.

C Expectation and Judgment Questionnaires

Tables 25 and 26 present the questionnaires used to measure participants' expectations toward the AI system and their subsequent judgment scores, respectively.

Table 5: Two-Way ANOVA Results for Suggestion Quality Manipulation (Pre-Experiment Validation, Section 4.5.1)

Effect	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial η_p^2
Suggestion quality	4520.469	2	2260.235	3083.821	$< .001$	0.897
Evaluation modality	30.804	3	10.268	14.010	$< .001$	0.056
Suggestion quality × Evaluation modality	28.142	6	4.690	6.399	$< .001$	0.051

Note. $R^2 = .898$ (adjusted $R^2 = .897$). Results computed with $\alpha = .05$.

Table 6: Two-Way ANOVA Results for Expectation Priming Manipulation (Pre-Experiment Validation, Section 4.5.2)

Effect	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial η_p^2
Expectation level	4112.344	2	2056.172	5067.955	$< .001$	0.935
Evaluation modality	46.693	3	15.564	38.362	$< .001$	0.140
Expectation level × Evaluation modality	7.144	6	1.191	2.935	0.008	0.024

Note. $R^2 = .898$ (adjusted $R^2 = .897$). Results computed with $\alpha = .05$.

Table 7: Results of Four-Way ANOVA on Judgment Scores in Experiment 1 (Section 5.4.1)

Effect	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial η_p^2
Suggestion quality	15153.594	95	159.512	234.088	$< .001$	0.889
Expectation level	59805.339	1	59805.339	87766.058	$< .001$	0.969
Scenario	12824.705	2	6412.352	9410.312	$< .001$	0.871
Evaluation modality	1352.994	3	450.998	661.853	$< .001$	0.416
Suggestion quality × Expectation level	7.200	1	7.200	10.566	0.001	0.004
Suggestion quality × Scenario	18.375	3	6.125	8.989	$< .001$	0.010
Suggestion quality × Evaluation modality	681.062	6	113.510	166.580	$< .001$	0.264
Expectation level × Scenario	7.965	2	3.982	5.844	0.003	0.004
Expectation level × Evaluation modality	92.073	6	15.345	22.520	$< .001$	0.046
Scenario × Evaluation modality	1.522	3	0.507	0.745	0.525	0.001
Suggestion quality × Expectation level × Scenario	29.047	9	3.227	4.736	$< .001$	0.015
Suggestion quality × Expectation level × Evaluation modality	11.258	3	3.753	5.507	$< .001$	0.006
Suggestion quality × Scenario × Evaluation modality	4.130	6	0.688	1.010	0.417	0.002
Expectation level × Scenario × Evaluation modality	45.772	18	2.543	3.732	$< .001$	0.024
Suggestion quality × Expectation level × Scenario × Evaluation modality	52.669	6	8.778	12.882	$< .001$	0.027

Note. $R^2 = .889$ (adjusted $R^2 = .885$). Results computed with $\alpha = .05$.

Table 8: Two-Way ANOVA Results on Absolute Judgment Shifts ($|\Delta J_1|$) by EV Magnitude and Evaluation Modality (Section 5.4.5)

Effect	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial η_p^2
EV magnitude	341.795	2	170.897	212.223	$< .001$	0.165
Evaluation modality	34.010	3	11.337	14.078	$< .001$	0.019
EV magnitude × Evaluation modality	8.456	6	1.409	1.750	0.106	0.005

Note. $R^2 = .181$ (adjusted $R^2 = .177$). Results computed with $\alpha = .05$.

Table 9: Three-Way ANOVA Results on Judgment Distortion ($|\Delta J_1|$) by Violation Direction, EV Magnitude, and Evaluation Modality (Section 5.4.6)

Effect	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial η_p^2
EV magnitude	14.115	1	14.115	23.174	$< .001$	0.016
Violation direction	650.389	1	650.389	1067.822	$< .001$	0.429
Evaluation modality	19.110	3	6.370	10.459	$< .001$	0.022
EV magnitude × Violation direction	52.840	1	52.840	86.753	$< .001$	0.057
EV magnitude × Evaluation modality	2.664	3	0.888	1.458	0.224	0.003
Violation direction × Evaluation modality	28.733	3	9.578	15.725	$< .001$	0.032
EV magnitude × Violation direction × Evaluation modality	7.063	3	2.354	3.865	0.009	0.008

Note. $R^2 = .459$ (adjusted $R^2 = .453$). Results computed with $\alpha = .05$.

Table 10: Three-Way ANOVA Results on Post-Evaluation Judgment Shifts (ΔJ_2) in Experiment 2 (Section 6.4.1)

Effect	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial η_p^2
Expectation level	815.156	2	407.578	672.003	$< .001$	0.388
Suggestion quality	889.979	2	444.989	733.685	$< .001$	0.409
Evaluation modality	54.554	3	18.181	29.977	$< .001$	0.041
Expectation level × Suggestion quality	264.991	4	66.248	109.227	$< .001$	0.171
Expectation level × Evaluation modality	49.314	6	8.219	13.551	$< .001$	0.037
Suggestion quality × Evaluation modality	34.158	6	5.693	9.387	$< .001$	0.026
Expectation level × Suggestion quality × Evaluation modality	151.950	12	12.663	20.878	$< .001$	0.106

Note. $R^2 = .637$ (adjusted $R^2 = .631$). Results computed with $\alpha = .05$.

Table 11: Three-Way ANOVA on Judgment Scores (J) Under CoT Reasoning (Section 7.3.1)

Effect	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial η_p^2
Suggestion quality	10510.501	2	5255.250	10387.262	$< .001$	0.907
Expectation level	940.422	3	313.474	619.597	$< .001$	0.467
Evaluation modality	0.112	2	0.056	0.111	0.895	0.000
Expectation level × Suggestion quality	544.692	6	90.782	179.435	$< .001$	0.336
Suggestion quality × Evaluation modality	0.655	4	0.164	0.323	0.862	0.001
Expectation level × Evaluation modality	2.714	6	0.452	0.894	0.498	0.003
Expectation level × Suggestion quality × Evaluation modality	2.631	12	0.219	0.433	0.951	0.002

Note. $R^2 = .918$ (adjusted $R^2 = .916$). Results computed with $\alpha = .05$.

Table 12: Three-Way ANOVA Results on Post-Evaluation Judgment Shifts (ΔJ_2) in Experiment 2 (Section 7.3.2)

Effect	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial η_p^2
Evaluation modality	1.209	1	1.209	3.612	0.057	0.001
Suggestion quality	479.457	2	239.729	715.831	$< .001$	0.308
Expectation level	572.139	2	286.070	854.205	$< .001$	0.347
Suggestion quality × Evaluation modality	0.401	2	0.201	0.599	0.549	0.000
Expectation level × Evaluation modality	1.111	2	0.556	1.659	0.191	0.001
Expectation level × Suggestion quality	601.487	4	150.372	449.011	$< .001$	0.358
Expectation level × Suggestion quality × Evaluation modality	0.792	4	0.198	0.591	0.669	0.001

Note. $R^2 = .606$ (adjusted $R^2 = .603$). Results computed with $\alpha = .05$.

Table 13: Social Issue 1: Suggestion Generation from GPT-4 under Different Prompted Capability Levels

Prompt Capability	Suggestions to Address "Climate Change"
Low capability	To combat climate change, it is recommended to use more wood products. Consideration could be given to eliminating the planting of trees, which take up space. New energy sources such as wind and solar power have a lot of uncertainties, so it may be safer to rely on traditional energy sources.
Moderate capability	To address climate change, we need to plant more trees. There is no need for waste segregation. However, care needs to be taken to conserve water. Painting city roofs white to reflect sunlight cools cities. Encouraging public transport, cycling, and walking can reduce dependence on fossil fuels.
High capability	Extensive tree-planting activities can significantly enhance carbon sequestration, thereby cleaning up the atmosphere. Enhanced recycling programs can reduce the need for landfill waste and raw material extraction, thereby reducing the overall carbon footprint. Encouraging the use of public transport, cycling, and walking can reduce dependence on fossil fuels.

Table 14: Social Issue 2: Suggestion Generation from GPT-4 under Different Prompted Capability Levels

Prompt Capability	Suggestions to Address "Rumor Mongering"
Low capability	When rumors start to emerge, although their impact may gradually dissipate, it is still possible to resist their spread by establishing regularly updated official communication channels. The impact of rumors is not considered significant and may not require serious concern, but some degree of control can still be exercised. In addition, the story can be relayed based on personal understanding.
Moderate capability	When rumors start to emerge, although their impact may gradually dissipate, it is still possible to resist their spread by establishing regularly updated official communication channels. The impact of rumors is not considered significant and may not require serious concern, but some degree of control can still be exercised. In addition, the story can be relayed based on personal understanding.
High capability	Establish official communication channels that are regularly updated. Collaborate with influential community leaders to disseminate accurate information. Strengthen relevant legislation to penalize perpetrators of cyber violence. Educate the public on how to recognize, verify, and challenge unsubstantiated information.

Table 15: Social Issue 3: Suggestion Generation from GPT-4 under Different Prompted Capability Levels

Prompt Capability	Suggestions to Address "Epidemic Transmission"
Low capability	Indoor gatherings can be held during an outbreak to strengthen community ties. International travel is possible during an outbreak as long as personal protective equipment is used. Public health bulletins and official guidelines should be treated as informational only, as they are often overly cautious and may unnecessarily restrict personal freedom.
Moderate capability	Indoor parties can be organized during an outbreak provided that participants wear masks and avoid sharing cutlery and cups. Vaccination may be taken or declined, and individuals are advised to consult healthcare providers for guidance. International travel remains possible with appropriate personal protective equipment. Public health bulletins and official guidelines are considered informational and may be overly cautious.
High capability	Sanitary and public health measures are essential for preventing epidemic transmission. Avoiding close contact with individuals who are visibly ill can substantially reduce risk. Vaccination is one of the most effective strategies for preventing infectious diseases. Public places should be equipped with adequate sanitary facilities, and frequently touched surfaces should be cleaned regularly using appropriate disinfectants.

Table 16: Social Issue 4: Suggestion Generation from GPT-4 under Different Prompted Capability Levels

Capability	Suggestions to Address "Increased Employment Pressure"
Low capability	To cope with increased employment pressure, consider adopting avoidance strategies and embracing a casual lifestyle, as stress is just a concept. There is no need to be overly anxious because most layoffs are due to broader societal changes rather than individual shortcomings. If employment pressures continue to rise, one possible option is to consider living in seclusion.
Moderate capability	There is no need to be overly anxious, and individuals can communicate more with friends to seek solutions, as many layoffs are driven by broader economic conditions. Layoffs may provide an opportunity to take a break, and individuals should cherish this period as a hard-earned rest. Believing in one's own value, individuals can engage in further study and self-improvement after sufficient relaxation.
High capability	Strengthening support systems through community-based programs can provide resources and counseling for individuals facing employment pressure. Advocacy for economic policies that address job scarcity and create new employment opportunities is necessary. Governments should provide basic social security protections, and educational initiatives should be implemented to enhance personal resilience and diversify workforce skills.

Table 17: Social Issue 5: Suggestion Generation from GPT-4 under Different Prompted Capability Levels

Capability	Suggestions to Address "Cyber-Violence"
Low capability	To deal with cyber violence, people can respond with sharper and more caustic comments. This is because only then will the other person realize the seriousness of the problem and avoid the recurrence of cyber violence. People can also threaten cyber bullies behind the screen while publishing and spreading their behavior to alert others.
Moderate capability	When someone behaves badly on the Internet, we should report him or her and curb the spread of such remarks. At the same time, individuals should be able to defend themselves while developing a sense of online civility. In the face of cyber violence, problems should be addressed through communication conducted with goodwill.
High capability	To address cyber-violence, it is recommended that clear and enforceable policies be developed. Victims of cyber-violence should be provided with necessary emotional and legal assistance. Technological solutions should complement these efforts by designing algorithms and platforms that proactively detect and mitigate online attacks.

Table 18: Social Issue 6: Suggestion Generation from GPT-4 under Different Prompted Capability Levels

Capability	Suggestions to Address "Transnational Crime"
Low capability	To deal with transnational crime, one could organize a persuasive session for criminals to convince them to abandon their illegal behavior through goodwill. Instead of relying on trained law enforcement agencies, it would be better to create a team of volunteer detectives, who would be more enthusiastic to participate in solving cases. Eliminating the exchange of different currencies would make it difficult for criminals to transfer their illegal gains and eliminate transnational crime at its root.
Moderate capability	To deal with transnational crime, those criminals who commit less serious crimes can be considered for exemption and extradition to their home countries. A global reward could be offered to increase the arrest rate, and even private investigators could be recruited to apprehend the criminals.
High capability	In tackling transnational crime, the joint efforts of international law enforcement and legal agencies are crucial. The establishment of consistent legal frameworks and definitions of transnational crime can facilitate such cooperation. This can be achieved by standardizing investigative techniques and law enforcement training to effectively respond to crimes that cross multiple jurisdictions.

Table 19: Personal Challenge 1: Suggestion Generation from GPT-4 under Different Prompted Capability Levels

Capability	Suggestions on How to "Work Out Effectively"
Low capability	For an effective workout, start by replacing all water intake with energy drinks to ensure consistently high energy levels. Fitness requires persistence and hence one should make it a habit to work out every day, no matter what. Exercise in tough conditions and push your body to its limits.
Moderate capability	To stay healthy, walking around and eating colorful foods will make you a winner. Run outside and play tag with your friends, even if it's very hot or pouring rain outside. Join your school's sports teams; the more activities you participate in, the better. Getting enough sleep ensures that you have enough energy for the next day, so the more sleep you get, the better.
High capability	It is vital to establish a balanced routine that incorporates cardiovascular and strength training. Cardiovascular activities such as running, cycling, or swimming enhance heart health and endurance. Strength training builds muscle and improves metabolism through weight lifting or self-weighted exercises. Consistency is key; regular exercise, even for short periods, is more beneficial than sporadic strenuous workouts.

Table 20: Personal Challenge 2: Suggestion Generation from GPT-4 under Different Prompted Capability Levels

Capability	Suggestions on How to Have a "Healthy Diet"
Low capability	To eat healthily, one may consider adopting a diet structure that is color-themed according to the days of the week, for example, green food on Mondays and red food only on Tuesdays. It is believed that a single diet can lead to good health, so a diet consisting only of fruits and vegetables is beneficial. Consider replacing water intake with fruit juices and sodas.
Moderate capability	Eating healthy means eating lots of fruits, vegetables, but not staples. Eating too many sweets and drinking too much Coke Sprite probably won't help your body, but it doesn't seem to hurt either. Drinking water is very important, drink as much as you can during the day, even if you are not thirsty. To stay energized, you should eat as soon as you feel hungry and eat as much as you can.
High capability	A healthy diet involves a balanced approach that includes a variety of nutrients to support overall health. The focus is on consuming natural foods, including a variety of fruits and vegetables, to ensure a broad range of vitamins and minerals are consumed. Portion control is important to avoid overeating, even with healthy foods.

Table 21: Personal Challenge 3: Suggestion Generation from GPT-4 under Different Prompted Capability Levels

Capability	Suggestions on How to "Reduce the Reliance on Mobile Phones"
Low capability	To reduce the reliance on mobile phones, one may consider securing it to a helium balloon to ensure that it is always out of reach. The government can introduce a policy to limit the time during which Internet access is available. Consider using traditional ways of keeping in touch, such as using letters.
Moderate capability	To cut down on mobile phone use, try finding fun things to do that don't require a screen, such as playing outside with friends. People can also consider securing it to a helium balloon or locking the phone in a safe and swallowing the key, making it difficult for people to get to it easily. Keeping mobile phones at a distance during work or study can increase productivity and minimize distractions. The government can introduce policies that limit the time during which internet access is possible.
High capability	Engage in other activities to fill the time typically spent on mobile phones, such as reading, engaging in hobbies, or spending time outdoors. Face-to-face interactions rather than virtual communication can foster stronger personal connections and reduce screen time. Set personal goals for mobile phone use and monitor screen time through in-built tools or apps.

Table 22: Personal Challenge 4: Suggestion Generation from GPT-4 under Different Prompted Capability Levels

Capability	Suggestions on How to "Save Money"
Low capability	To save money, insist on not using the heating or air conditioning no matter what the weather is, which will greatly reduce utility bills. Stick to walking anywhere, even if the distance is very far, to avoid transport costs. Stick to eating cheap food, which will save on dietary costs. Try not to invest as any investment is risky.
Moderate capability	To save money, reduce the frequency of air-conditioning use and walk out of the house whenever possible. Keep meals simple, consider eating more instant noodles and compressed biscuits, and from time to time you can go to a restaurant, but limit the number of times. Sharing toys and games with friends means not having to buy new things. Refuse to invest in anything and hide all your savings under your mattress in the firm belief that this is the safest way to avoid bank charges or losses.
High capability	To effectively save money, individuals should first create a detailed budget to track all income and expenses. Prioritizing needs over wants in purchasing decisions helps focus spending on necessities. Regularly reviewing and adjusting budgets to reflect changes in financial goals or circumstances ensures continued adherence to the savings plan.

Table 23: Personal Challenge 5: Suggestion Generation from GPT-4 under Different Prompted Capability Levels

Capability	Suggestions on How to "Ensure Quality Sleep"
Low capability	Use electronic devices or play music to help you relax. Use a spontaneous sleep schedule to make sleep flexible. Turn your bed into a multi-functional activity center: eat, work, and play on it. Finally, exercising before bedtime speeds up sleep by helping your body feel tired faster through physical exertion.
Moderate capability	To get a superb night's sleep, think like a sleepy cat. Firstly, make your bed the cosiest place you can. And, put on your favourite pajamas. Before you go to bed, play more video games, but catch the sleepy feeling and fall asleep as soon as you're sleepy, whether it's day or night. Hunger is not conducive to sleep, so if you feel hungry before bed you should eat before going to bed.
High capability	The sleep environment should be comfortable, which includes a quiet, dark, cool room. Electronic devices emit blue light, which can interfere with sleep, so avoid screens at least an hour before bedtime. Caffeine and nicotine are stimulants that can impede sleep, so limiting your intake of these substances.

Table 24: Personal Challenge 6: Suggestion Generation from GPT-4 under Different Prompted Capability Levels

Capability	Suggestions on How to "Keep Your Mood Cheerful"
Low capability	To keep your mood happy, watch films that highlight the hardships of life, which will make your own troubles seem less important. Fill your living space with dull and dark colors to create an atmosphere of stillness. Being alone most of the time and being quiet is the key to happiness. Sugar intake contributes to happiness.
Moderate capability	To keep your mood happy, start by doing things that make you laugh. Eat your favourite snack. Shopping can also release stress. Being destructive to something can also release a depressing mood, venting your unhappiness out loud or crying. Or exercise so strenuously that you feel tired and go to sleep, and you won't feel sad in your sleep.
High capability	Regular physical activity reduces stress. Prioritizing sleep is essential for mood regulation and stress management. Engaging in hobbies and activities that bring joy can increase happiness and reduce feelings of stress. Maintaining relationships and spending time with loved ones can enhance emotional support and happiness.

Table 25: Expectation Questionnaire

Dimension	Items
Performance	1. I believe the AI can understand and solve complex problems. 2. I believe the AI to handle challenging or ambiguous tasks effectively. 3. I believe the AI can perform well across different situations.
Decision Support	4. I believe the AI can provide me with valuable decision support. 5. The AI will help me make better choices. 6. I expect the AI's suggestions to improve the quality of my decisions.
Accuracy	7. I expect the AI to provide highly accurate results. 8. The AI will deliver precise and correct answers. 9. I believe the AI's output will be free from major errors.

Table 26: Judgment Score Questionnaire

Dimension	Items
Acceptance	1. The suggestion is relevant to the task. 2. The suggestion is helpful. 3. The suggestion is appropriate for my needs.
Trust	4. I trust the content of the suggestion. 5. I trust the source that provided the suggestion. 6. I feel confident relying on this suggestion.